

GEMP ACCESS COURSE: BIOLOGICAL CHEMISTRY

Lecturer: Christopher Larbie, *PhD*

Lecture 6: Alkenes

Kwame Nkrumah University of Science and Technology, Kumasi | Leaders In Change

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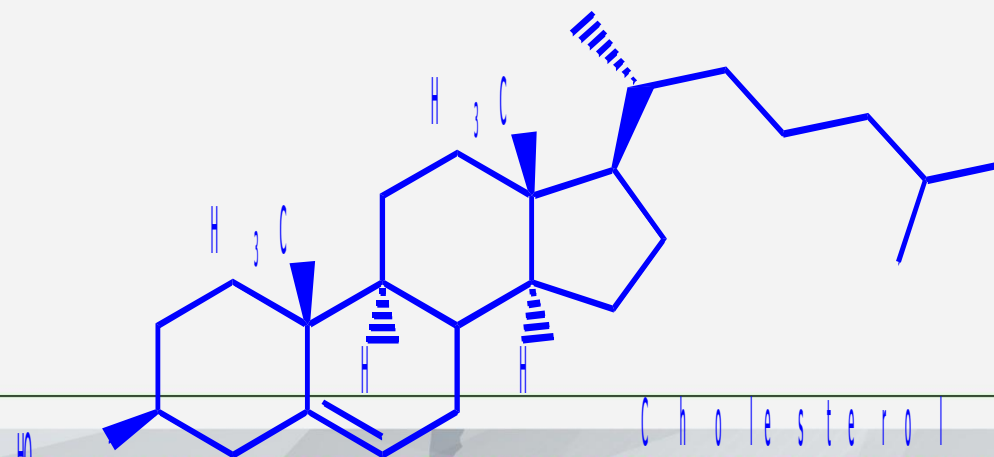
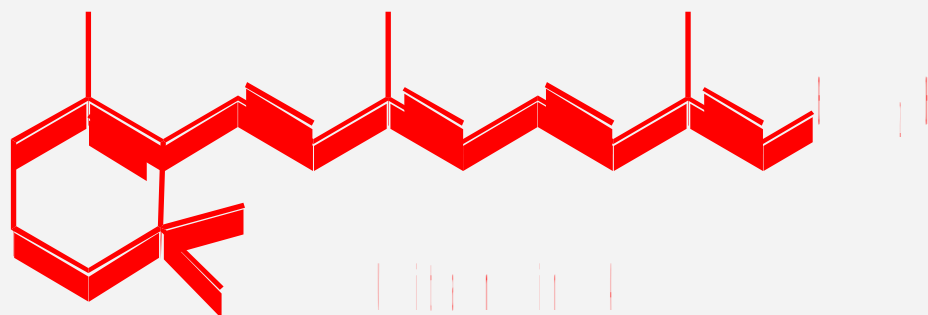
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Introduction

- ❖ Alkenes
 - Hydrocarbons containing C=C
 - Old name: olefins



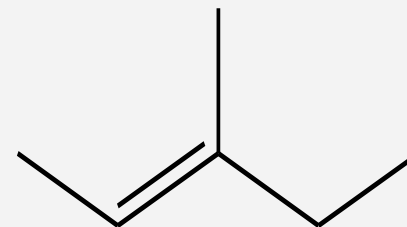
NOMENCLATURE



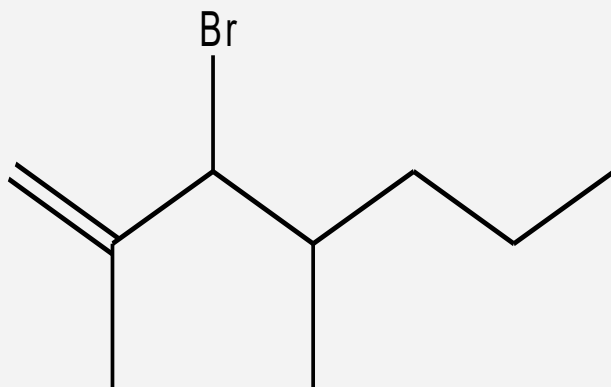
2-butene



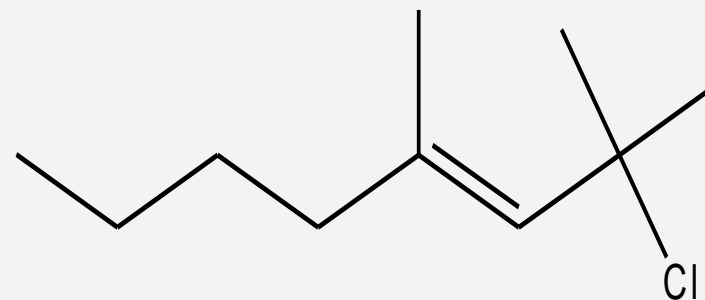
2-hexene



3-methyl-2-hexene



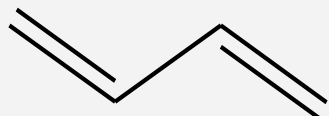
3-bromo-2,4-dimethyl-1-heptene



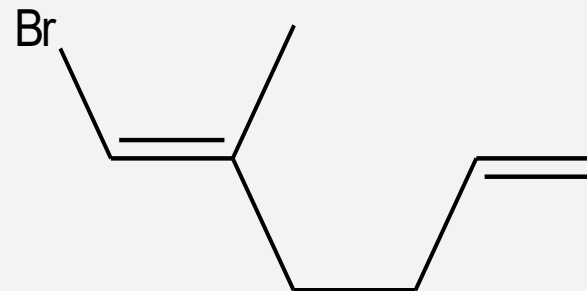
2-chloro-2,4-dimethyl-3-octene



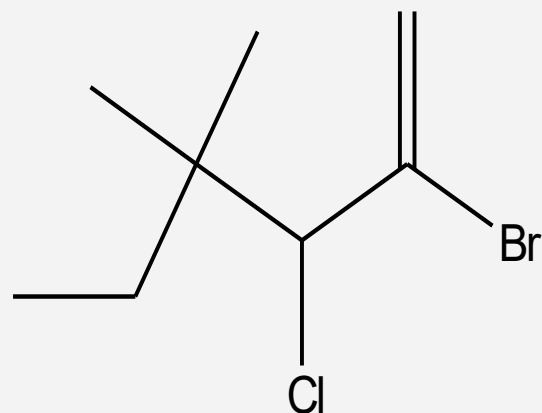
NOMENCLATURE OF ALKENES



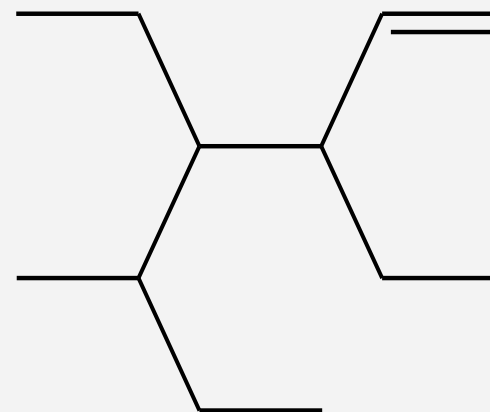
1,3-butadiene



1-bromo-2-methyl-1,5-hexadiene



2-bromo-3-chloro-4,4-dimethyl-1-hexene



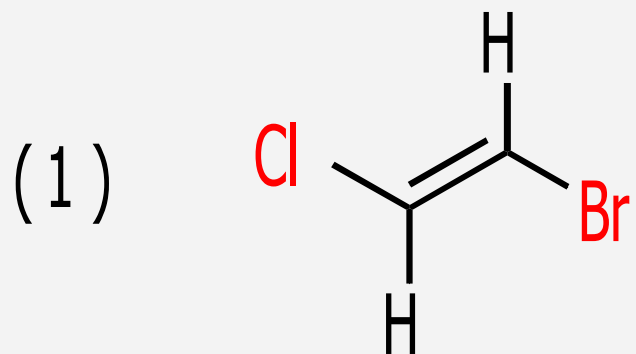
3,4-diethyl-5-methyl-1-heptene



The (E) - (Z) System for Designating Alkene Diastereomers

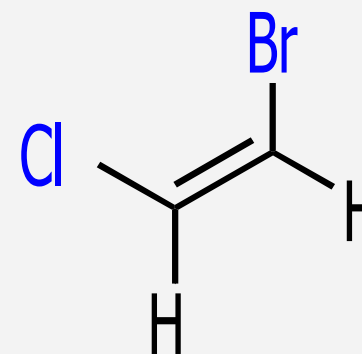
❖ Cis-Trans System

- Useful for 1,2-disubstituted alkenes
- Examples:



trans -1-Bromo-2-chloroethene

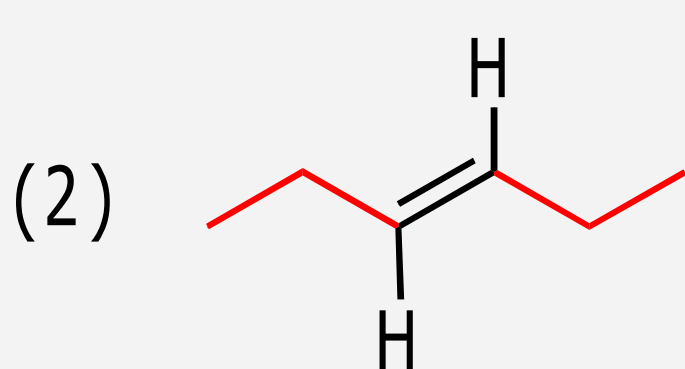
vs



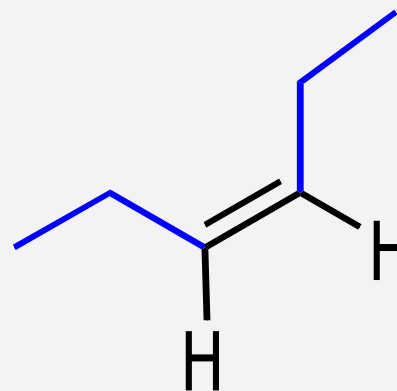
cis -1-Bromo-2-chloroethene



• Examples

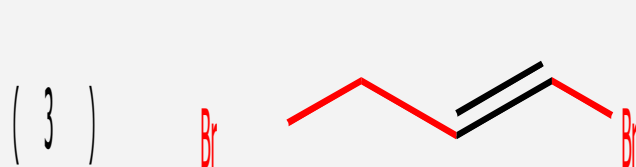


vs

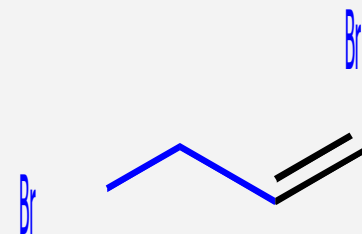


trans -3-Hexene

cis -3-Hexene



vs



trans - 1,3 -

cis - 1,3 -

Dibromopropene

Dibromopropene



❖ The Cahn-Ingold-Prelog (*E*) - (*Z*) Convention

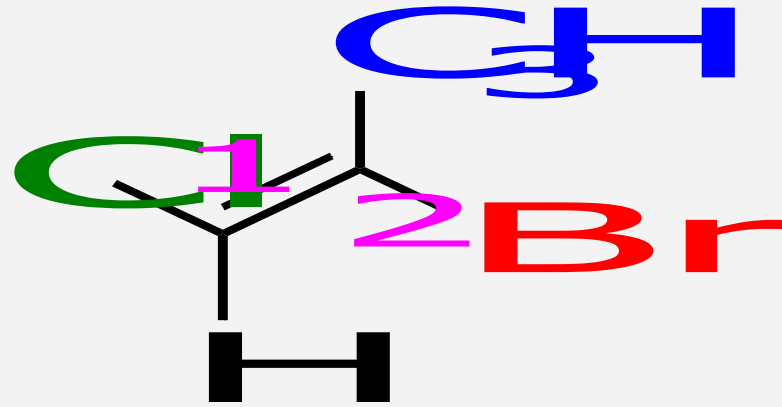
- The system is based on the atomic number of the attached atom
- The higher the atomic number, the higher the priority



- ❖ The Cahn-Ingold-Prelog (*E*) - (*Z*) Convention
 - (*E*) configuration – the highest priority groups are on the **opposite** side of the double bond
 - ◆ “*E*” stands for “**entgegen**”; it means “**opposite**” in German
 - (*Z*) configuration – the highest priority groups are on the **same** side of the double bond
 - ◆ “*Z*” stands for “**zusammen**”; it means “**together**” in German



- Examples



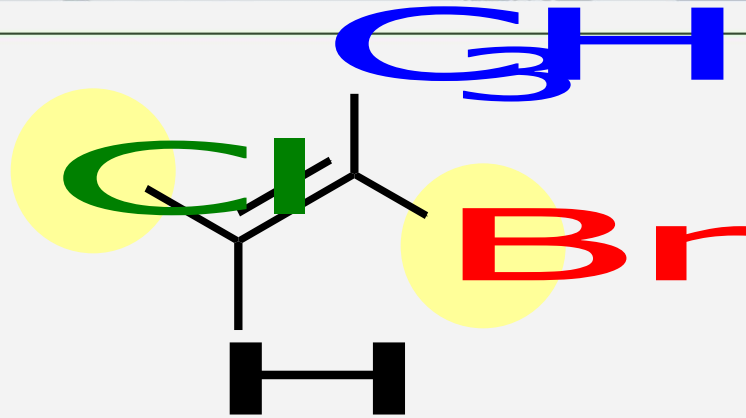
Priority 2: Priority of Br > CH₃

Priority 1: Priority of Cl > H

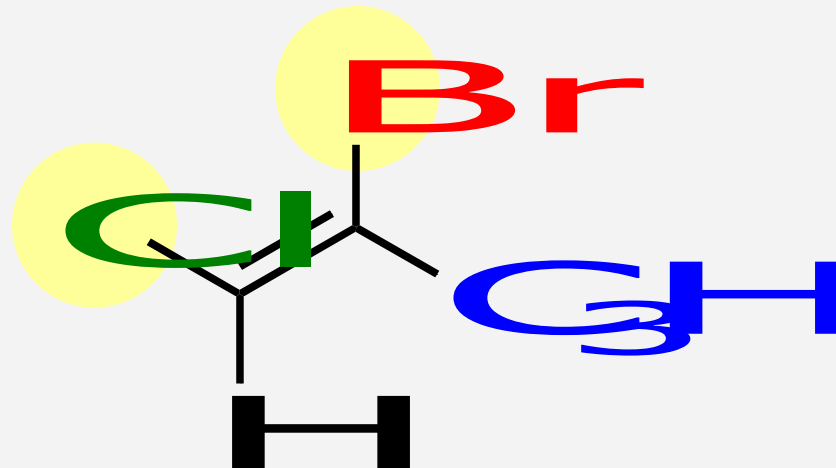
⇒ Highest priority groups
are Br (priority 2)
and Cl (priority 1)



- Examples



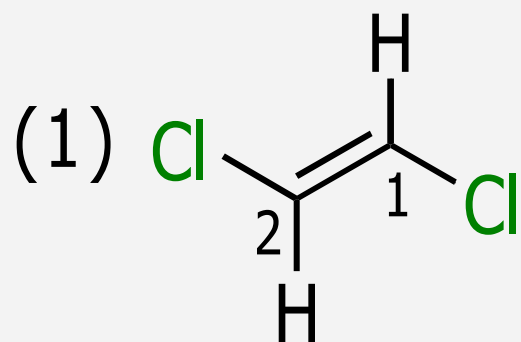
~~(E)-1-bromo-2-chloroethene~~



~~(Z)-1-bromo-2-chloroethene~~



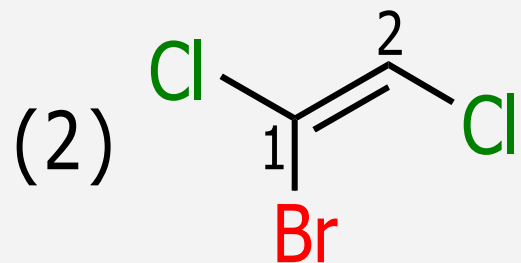
- Other examples



(*E*)-1,2-Dichloroethene
[or *trans*-1,2-Dichloroethene]

C1: Cl > H

C2: Cl > H



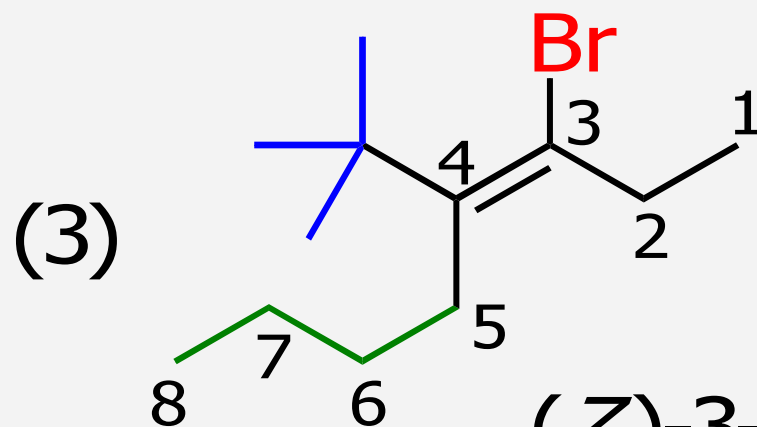
(*Z*)-1-Bromo-1,2-dichloroethene

C1: Br > Cl

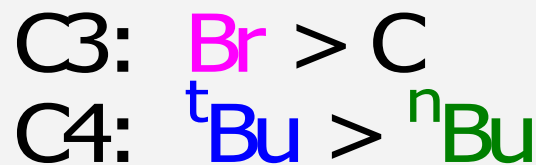
C2: Cl > H



- Other examples

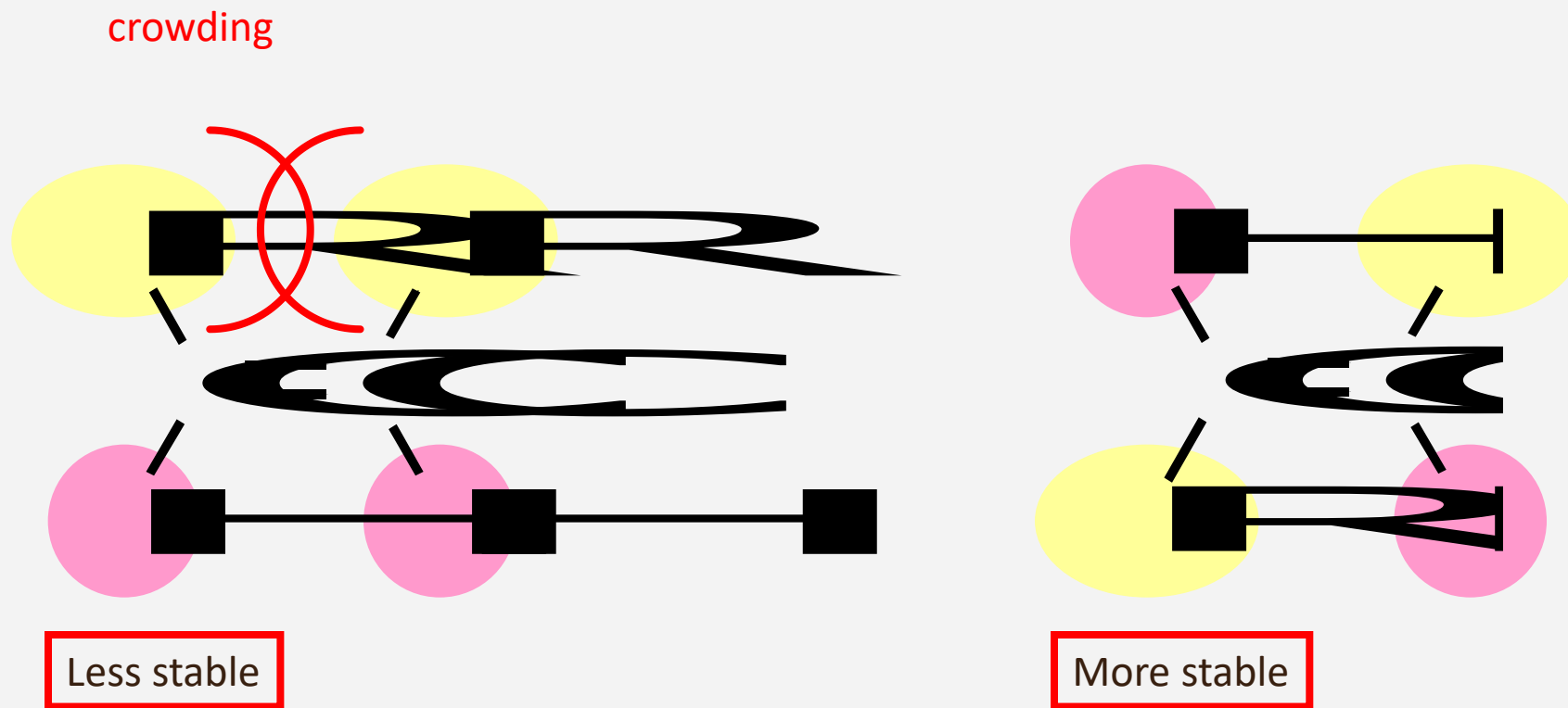


(*Z*)-3-Bromo-4-*tert*-butyl-3-octene



Relative Stabilities of Alkenes

- ❖ *Cis* and *trans* alkenes do not have the same stability

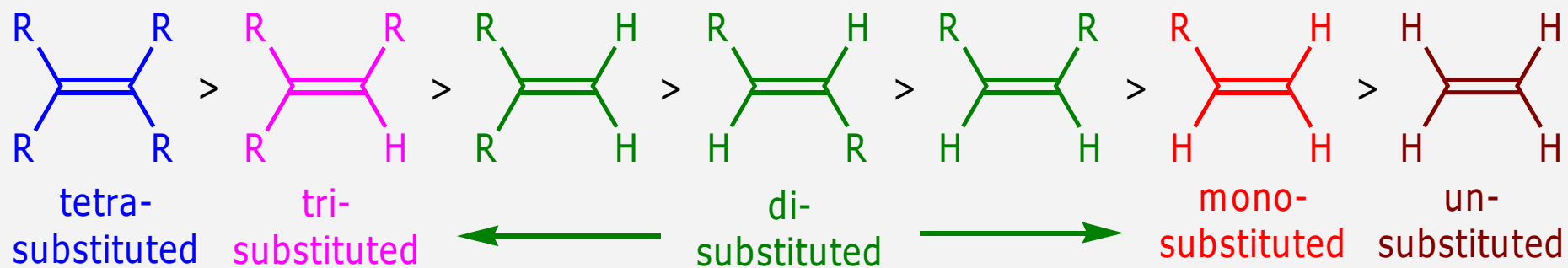


Overall Relative Stabilities of Alkenes

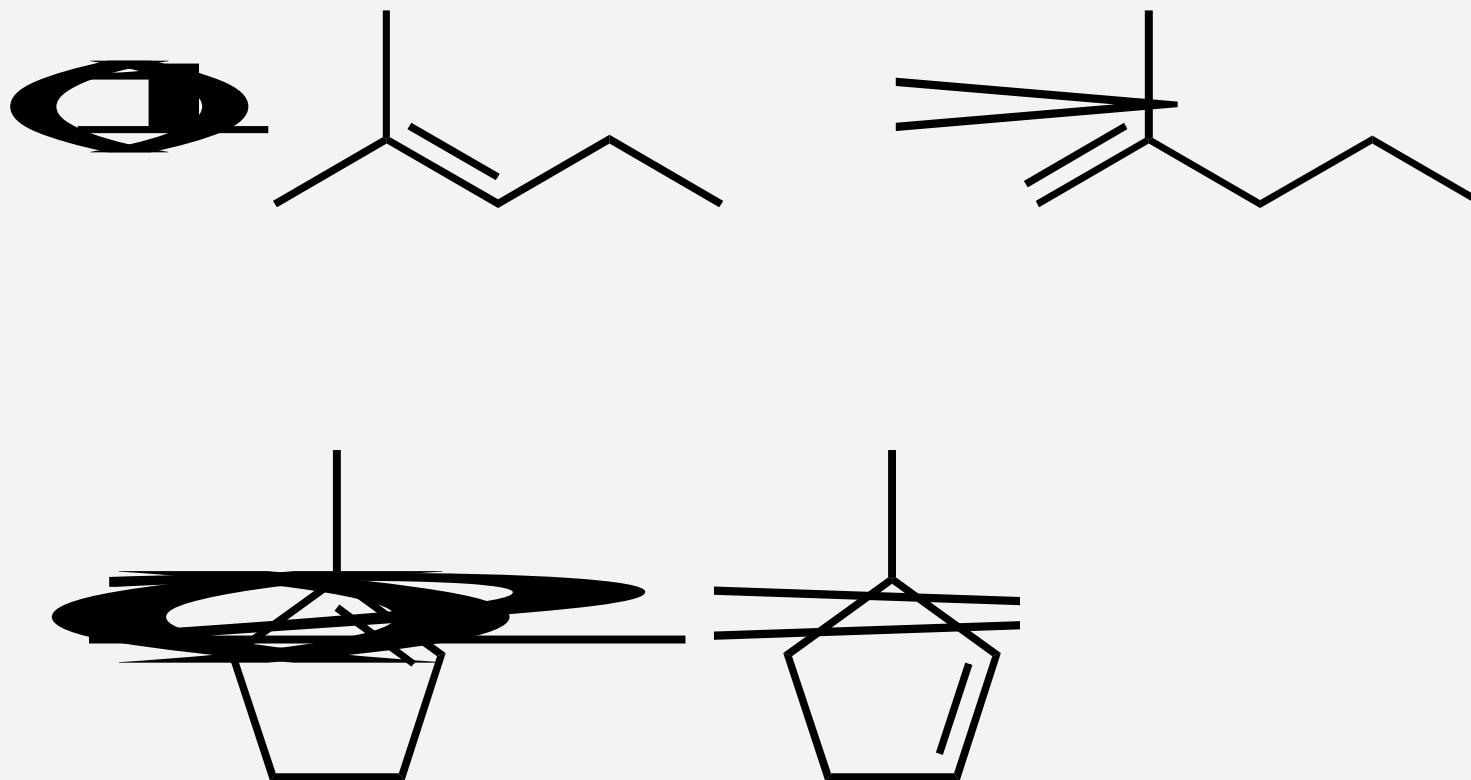
- ❖ The greater the number of attached alkyl groups (i.e., the more highly substituted the carbon atoms of the double bond), the greater the alkene's stability.



❖ Relative Stabilities of Alkenes

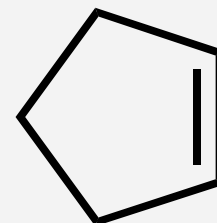
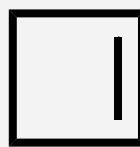


❖ Examples of stabilities of alkenes

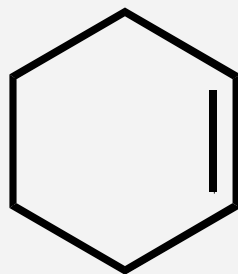


Cycloalkenes

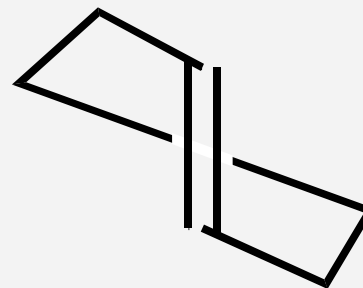
- ❖ Cycloalkenes containing 5 carbon atoms or fewer exist only in the *cis* form



- ❖ *Trans* – cyclohexene and *trans* – cycloheptene have a very short lifetime and have not been isolated



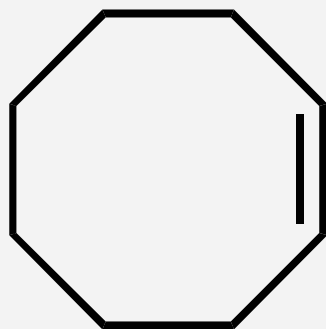
c y c l o h e x e n e



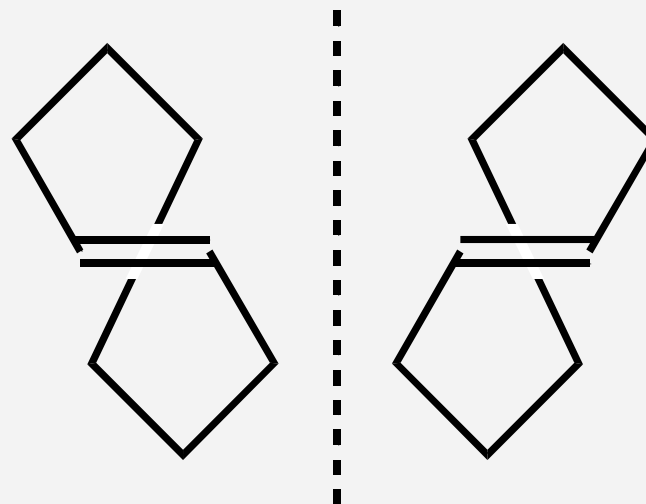
H y p o t h e t i c a l
t r a n s - c y c l o h e x e n e
(t o o s t r a i n e d t o e x i s t a t r . t .)



- ❖ *Trans* – cyclooctene has been isolated and is chiral and exists as a pair of enantiomers



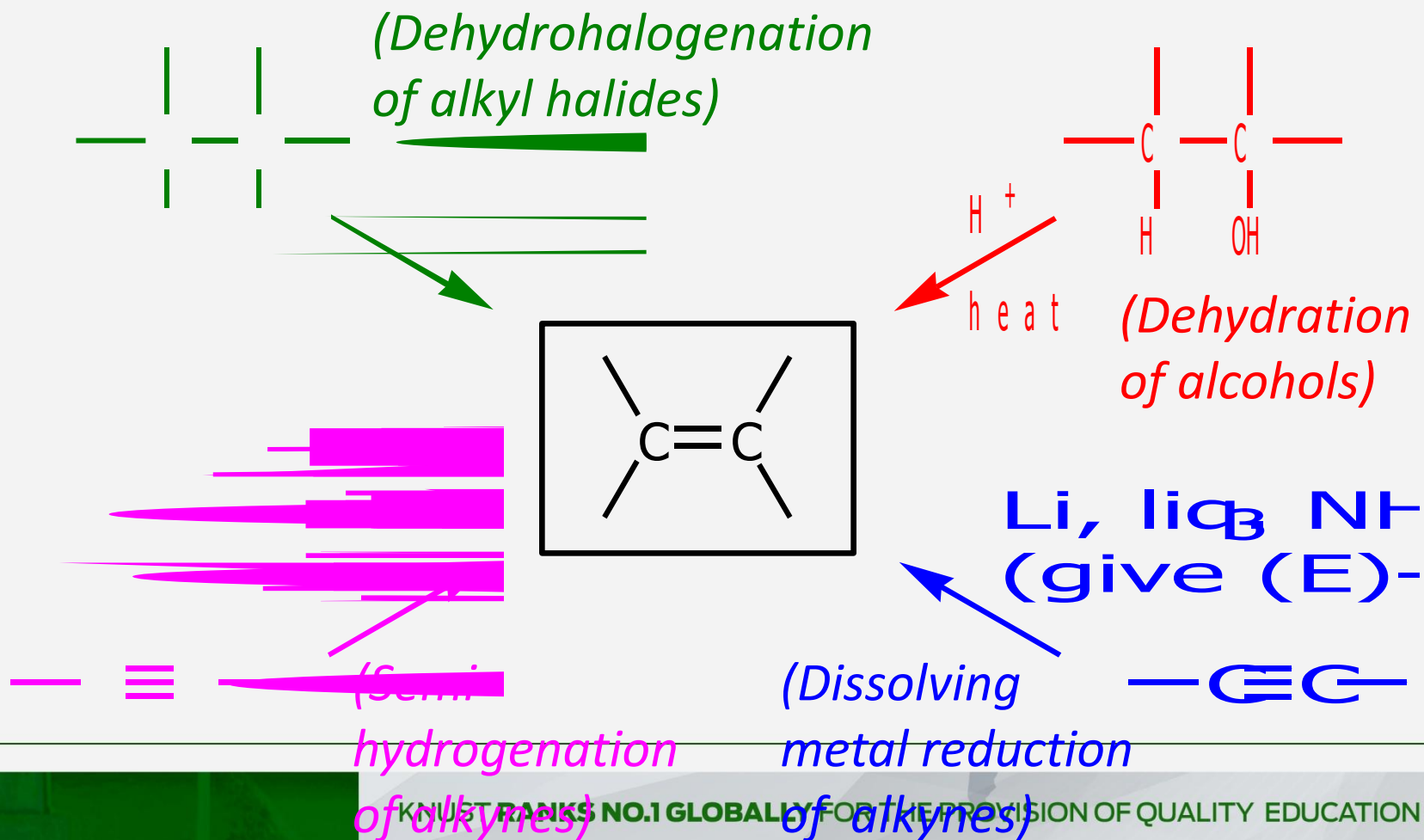
cis-cyclooctene



trans-cyclooctenes

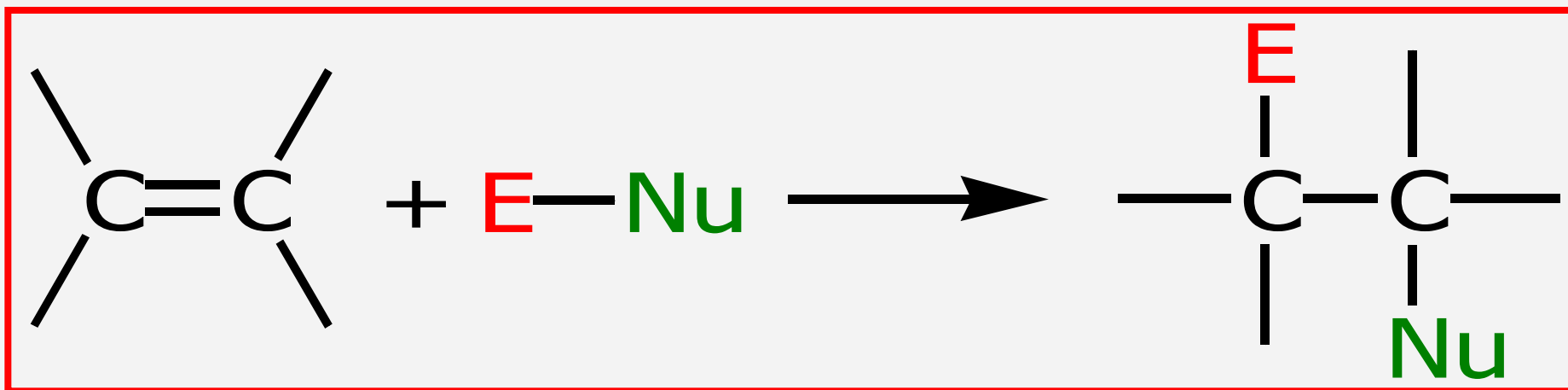


Summary of Methods for the Preparation of Alkenes



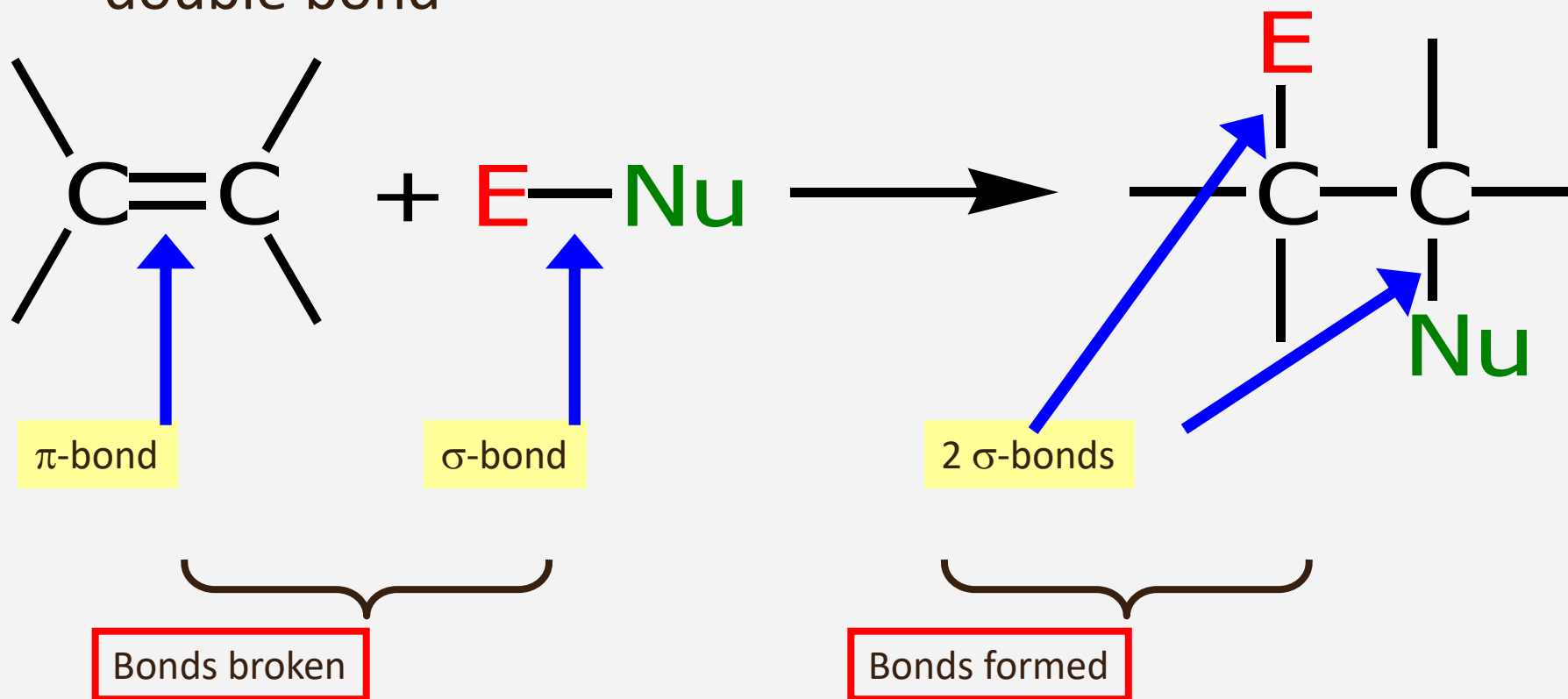
REACTIONS OF ALKENE

Addition Reactions of Alkenes

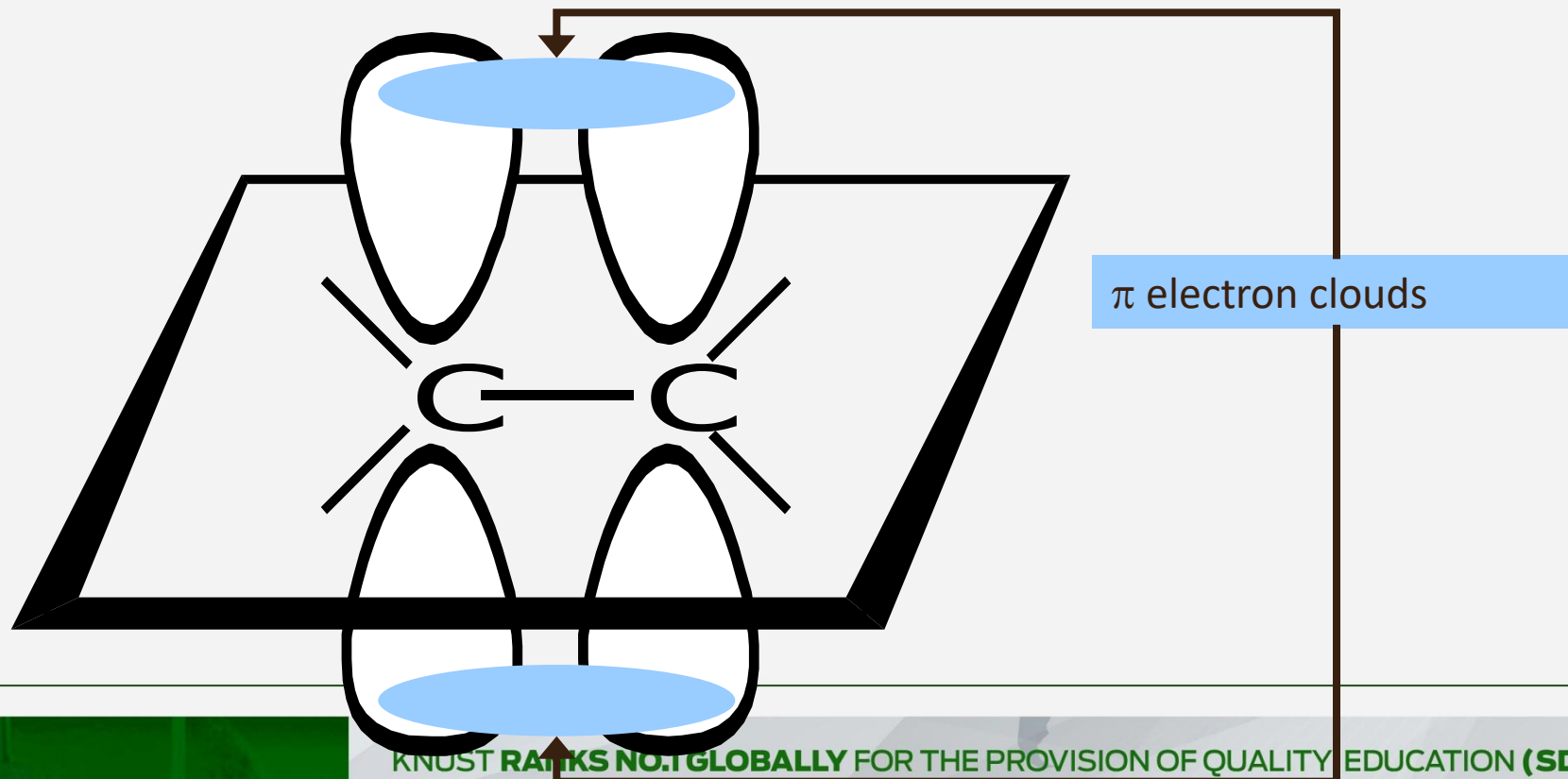


1A. How To Understand Additions to Alkenes

- ❖ This is an addition reaction: $E-Nu$ added across the double bond

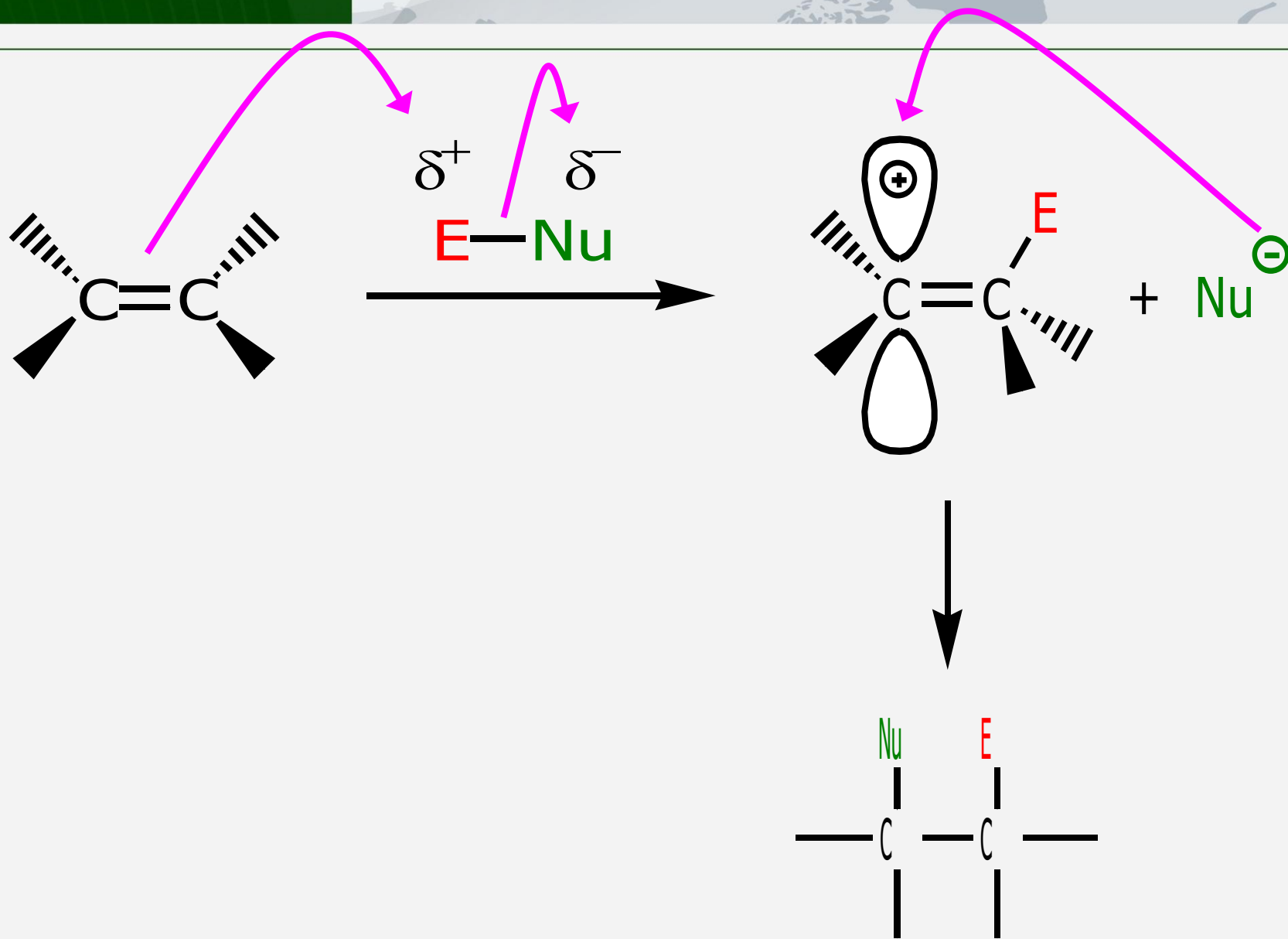


- ❖ Since p bonds are formed from the overlapping of π orbitals, π electron clouds are above and below the plane of the double bond



- ❖ Electrophilic
 - electron seeking
 - C=C and C≡C π bonds are particularly susceptible to electrophilic reagents (electrophiles)
- ❖ Common electrophile
 - H^+ , X^+ ($X = Cl, Br, I$), Hg^{2+} , etc.
- ❖ In an electrophilic addition, the π electrons seek an electrophile, breaking the π bond, forming a σ bond and leaving a positive charge on the vacant π orbital on the adjacent carbon. Addition of B^- to form a σ bond provides an addition product

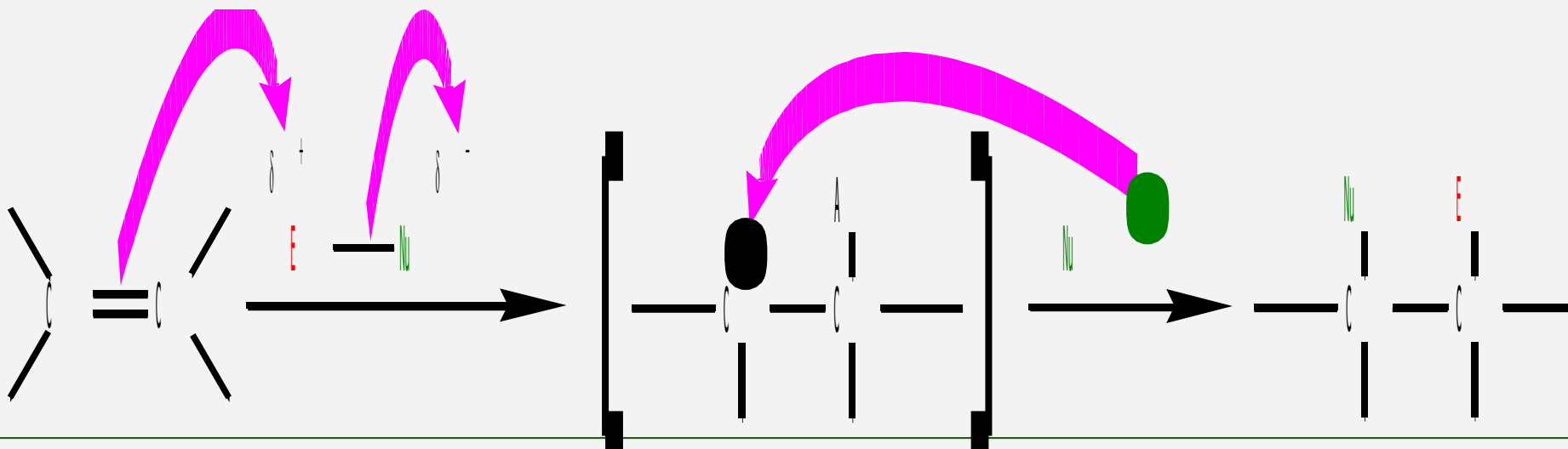




1. Electrophilic Addition of Hydrogen Halides to Alkenes:

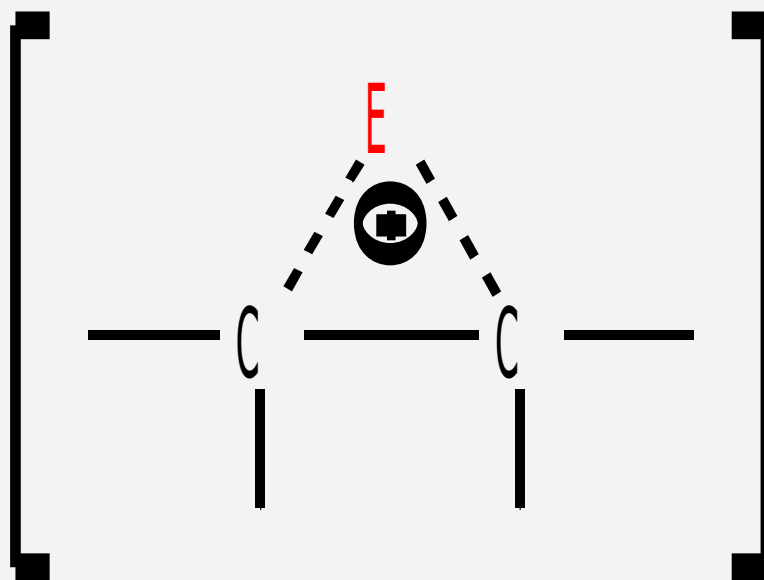
Mechanism and Markovnikov's Rule

❖ Mechanism



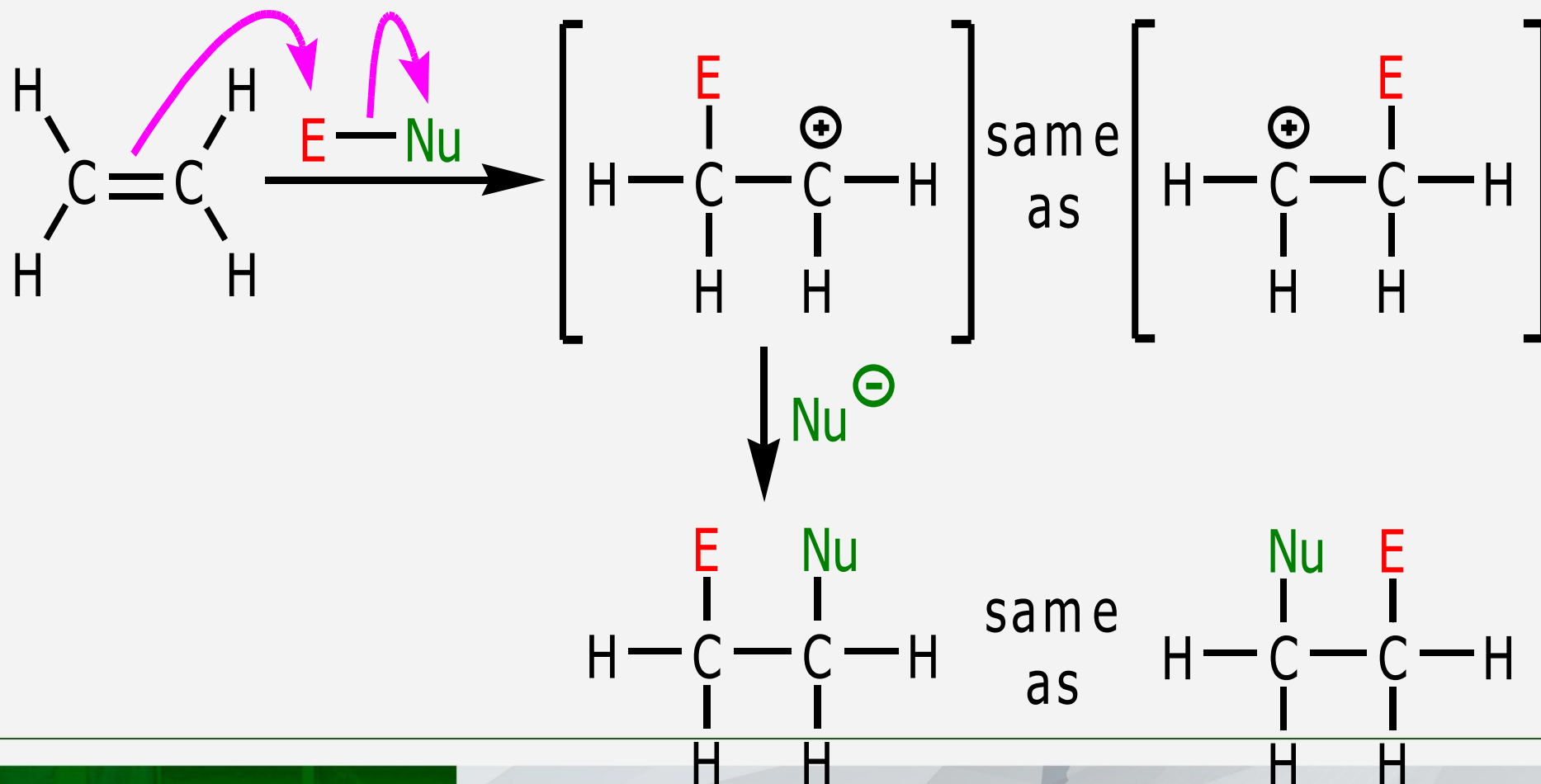
❖ Mechanism

- Sometimes do not go through a “free carbocation”, may go via



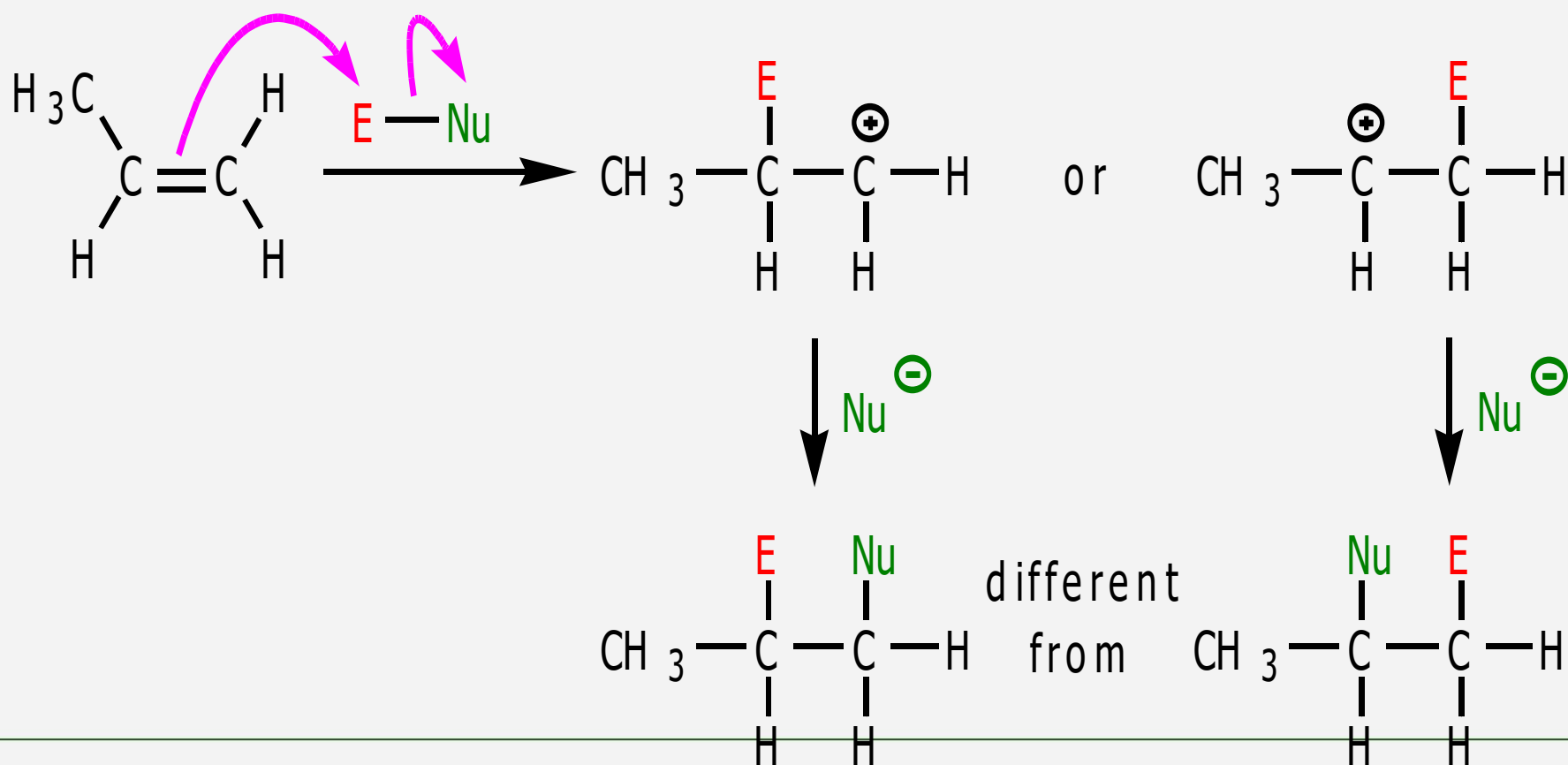
❖ Markovnikov's Rule

- For symmetrical substrates, no problem for regiochemistry



❖ Markovnikov's Rule

- But for unsymmetrical substrates, two regioisomers are possible



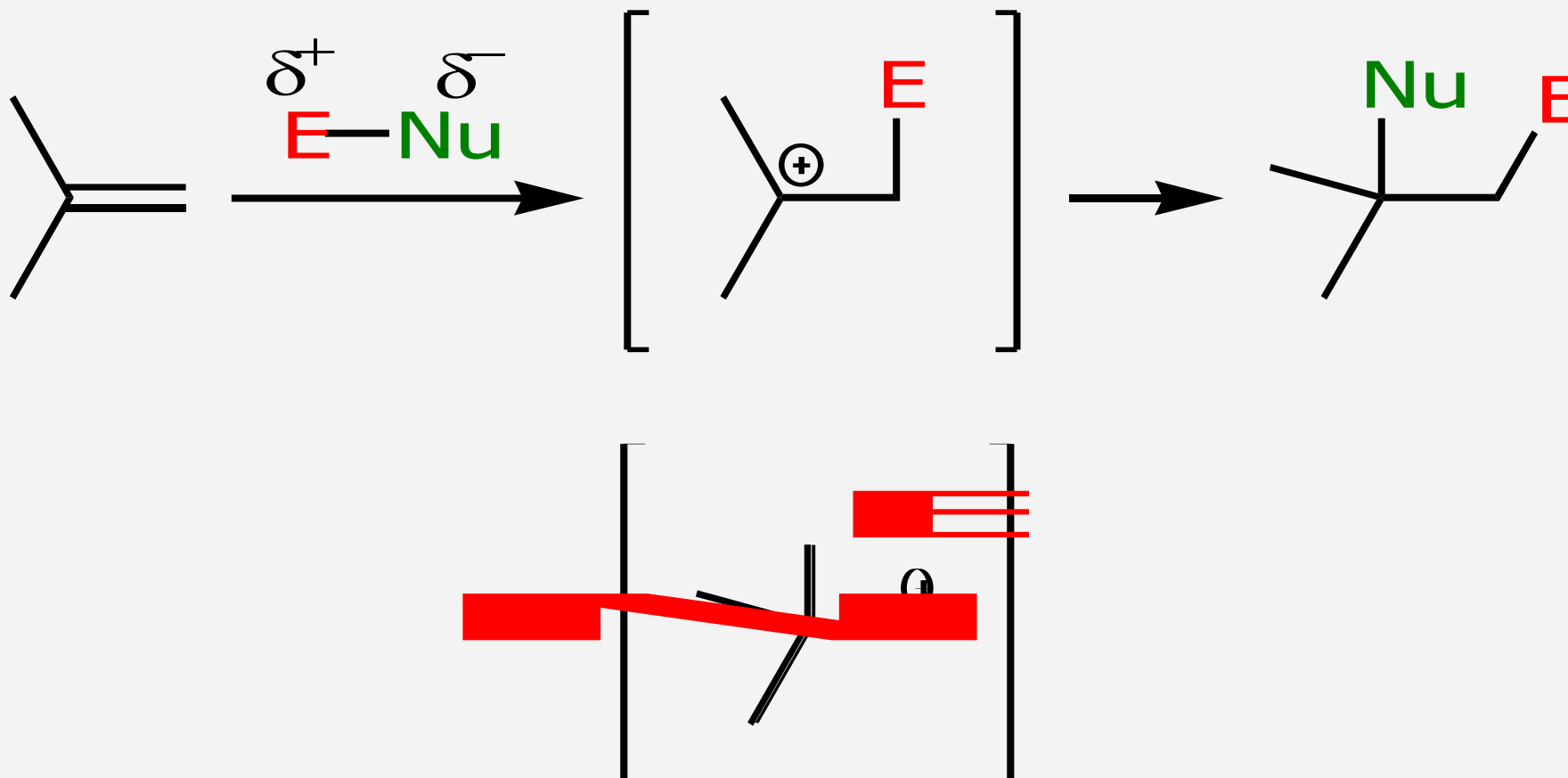
❖ Markovnikov's Rule

- In the electrophilic addition of an unsymmetrical electrophile across a double bond of an alkene, the more highly substituted and more stabilized carbocation is formed as the intermediate in preference to the less highly substituted and less stable one



❖ Markovnikov's Rule

● Thus

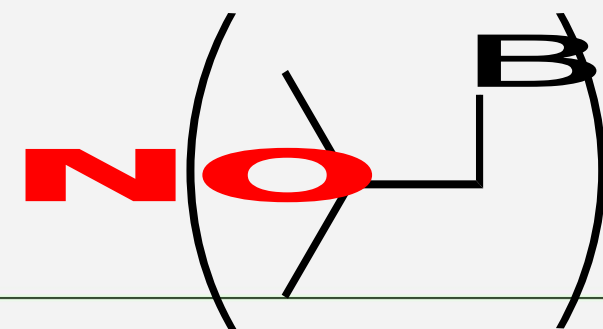
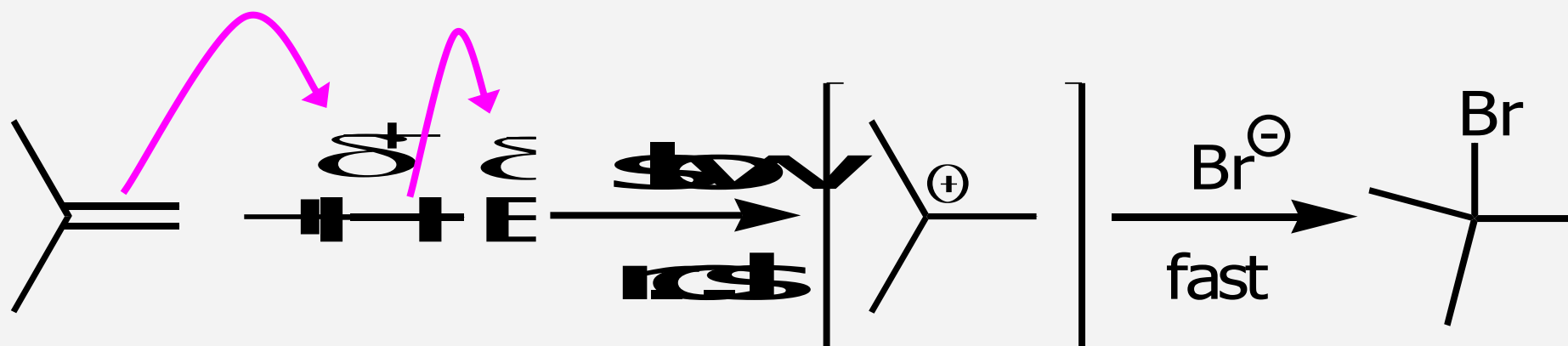


Note: carbocation stability $\Rightarrow 3^\circ > 2^\circ > 1^\circ$

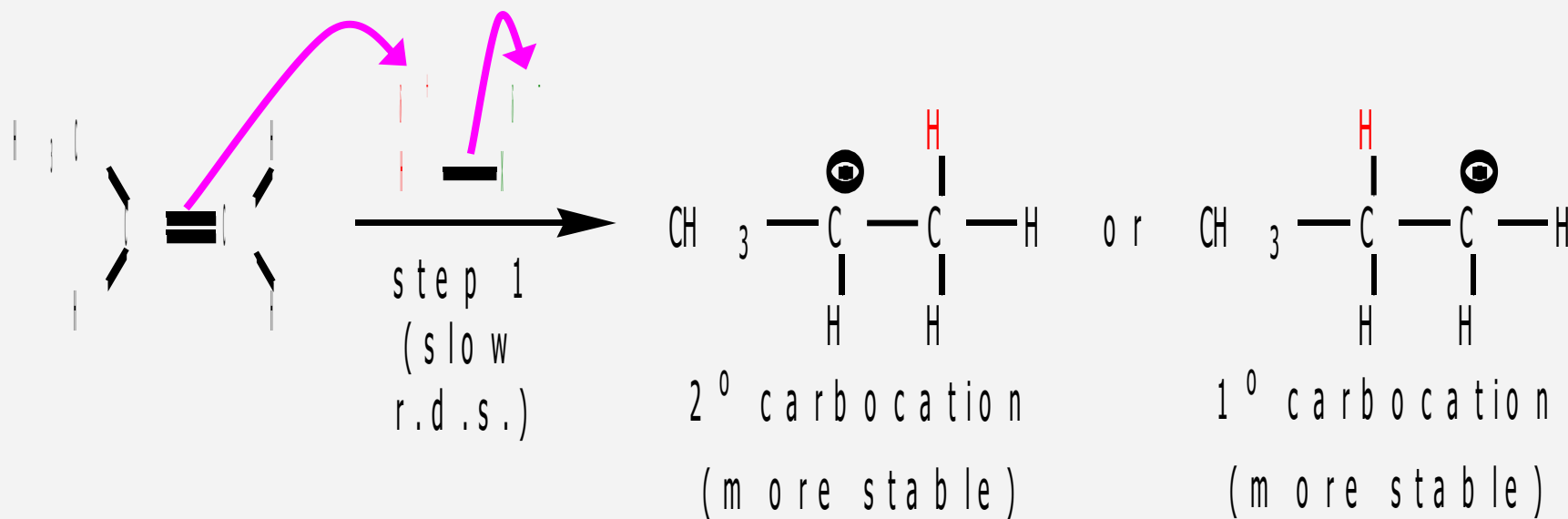


❖ Addition of Hydrogen Halides

- Addition of HCl, HBr and HI across a C=C bond
- H^+ is the electrophile

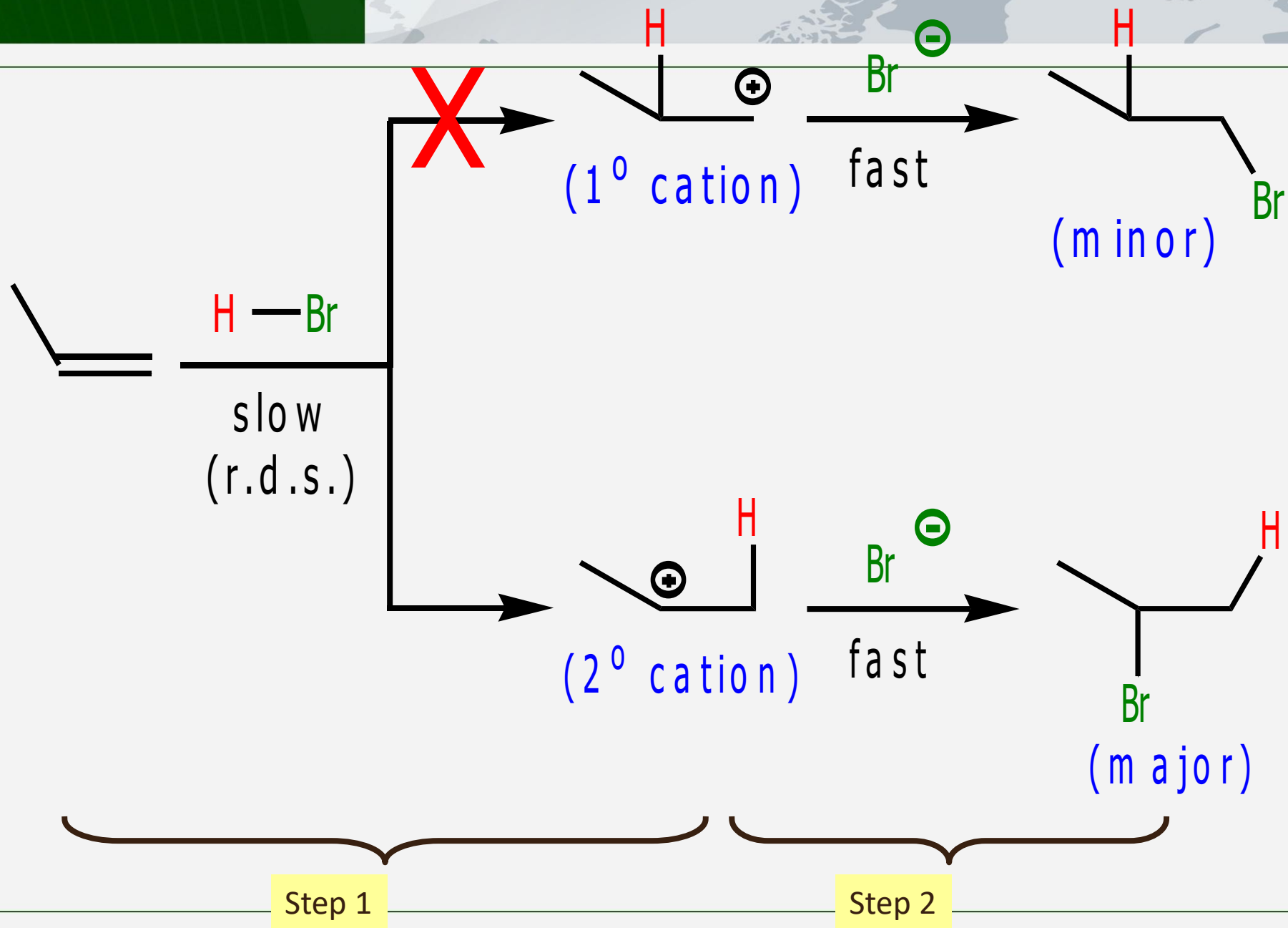


Theoretical Explanation of Markovnikov's Rule

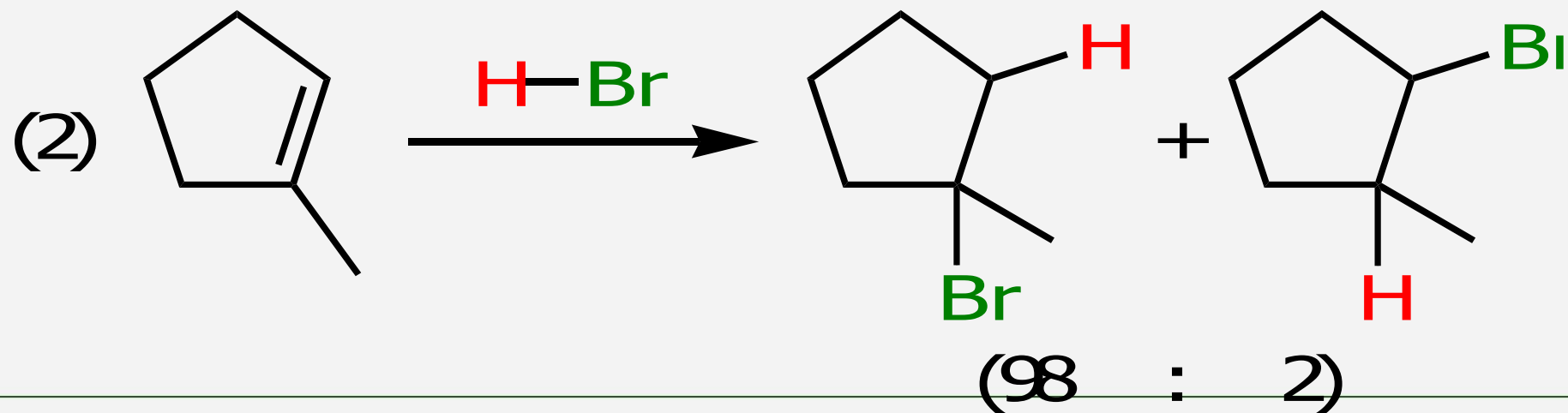
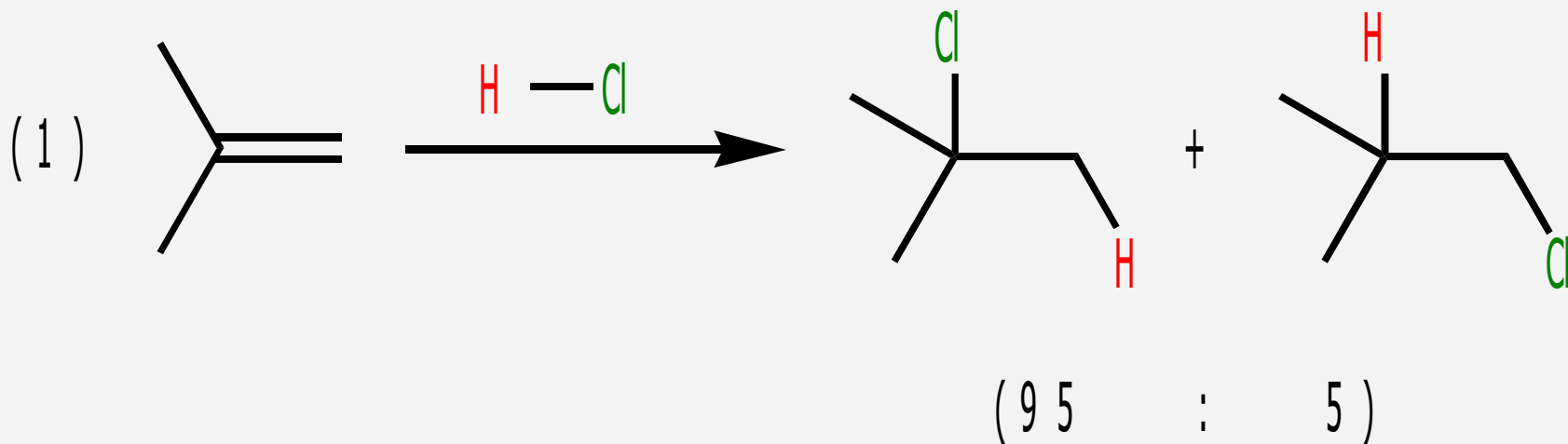


- ❖ One way to state Markovnikov's rule is to say that in the addition of HX to an alkene, the hydrogen atom adds to the carbon atom of the double bond that already has the greater number of hydrogen atoms





❖ Examples

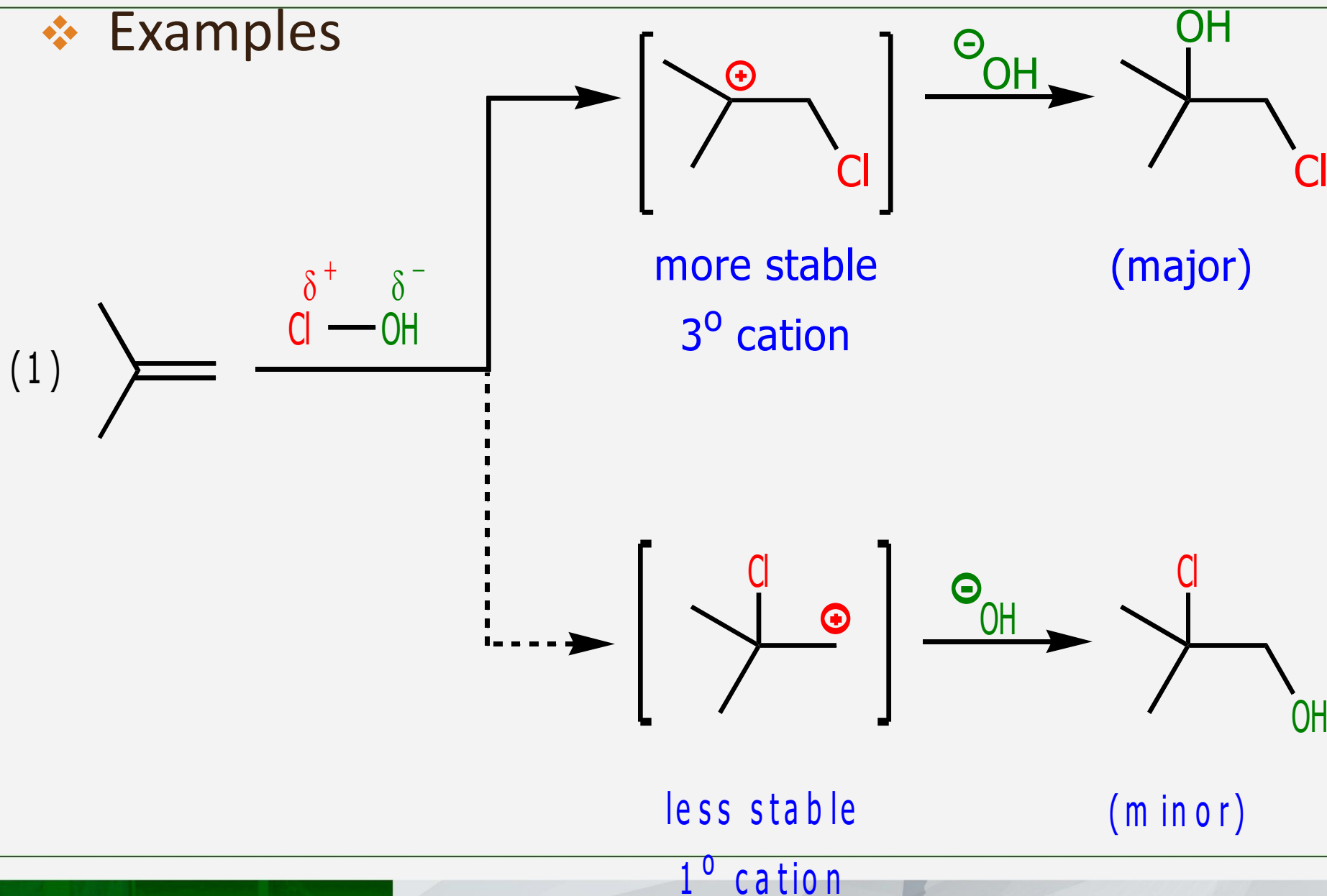


2B. Modern Statement of Markovnikov's Rule

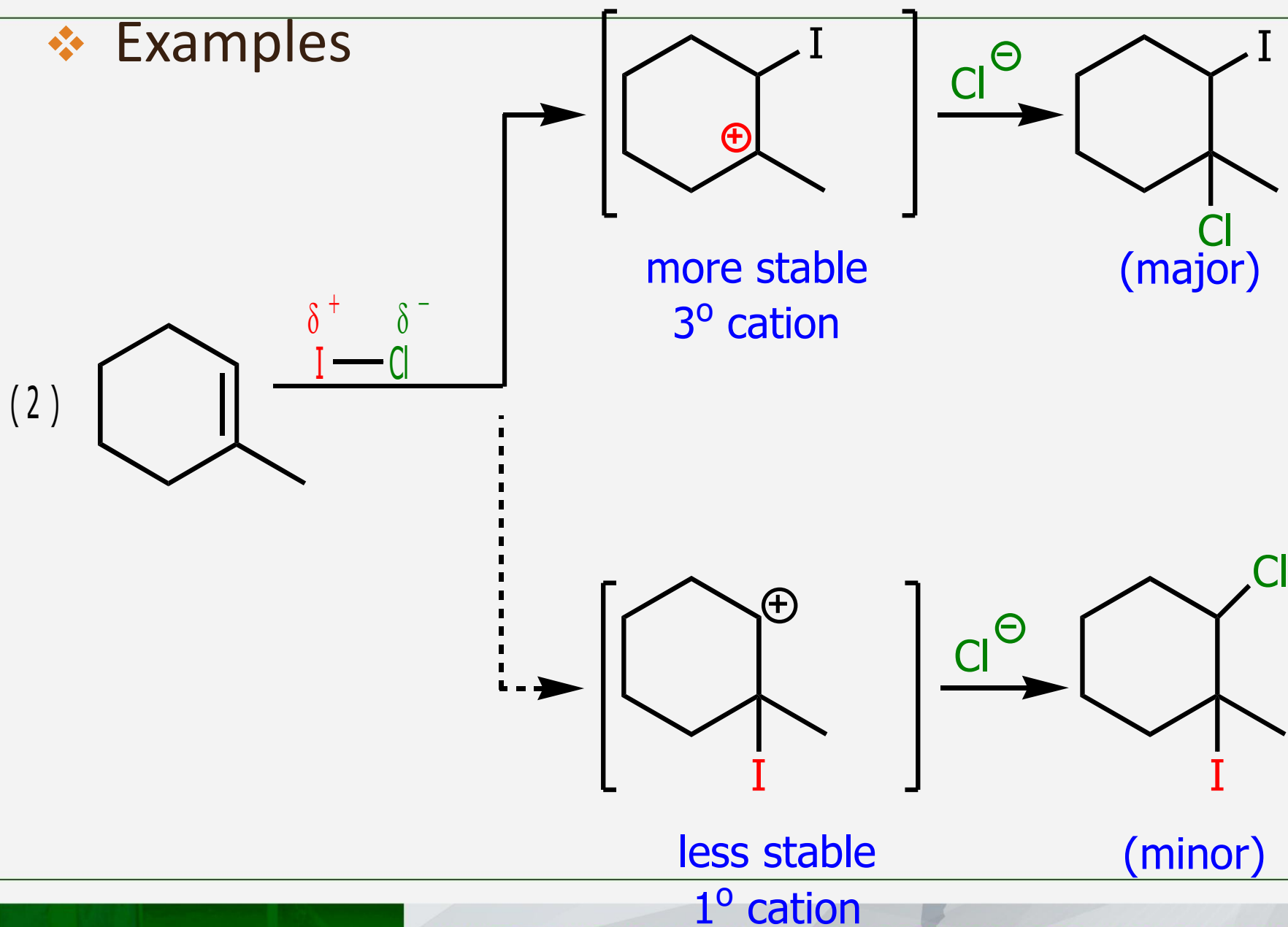
- ❖ In the ionic addition of an unsymmetrical reagent to a double bond, the positive portion of the added reagent attaches itself to a carbon atom of the double bond so as to yield the more stable carbocation as an intermediate



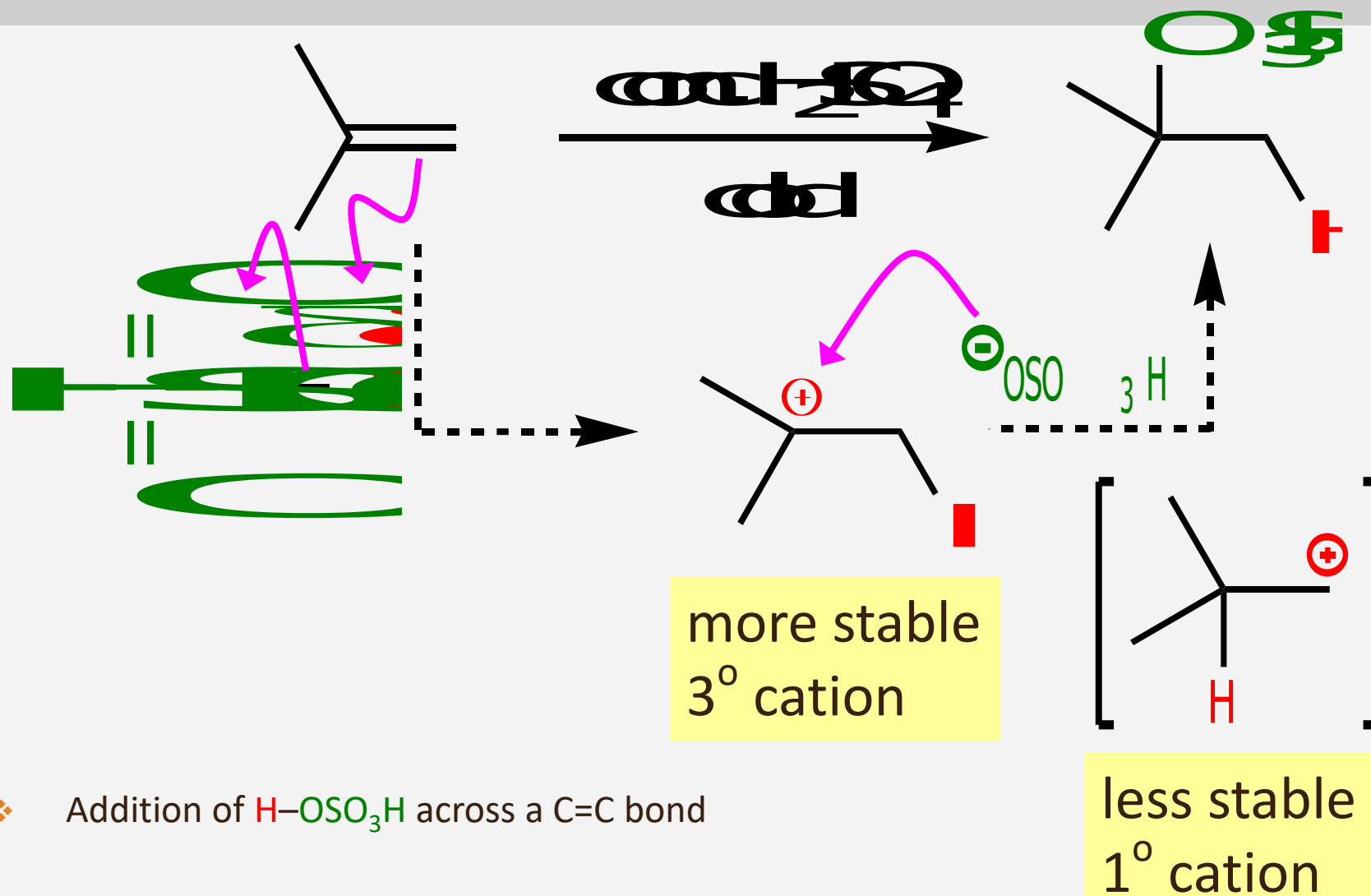
❖ Examples



❖ Examples



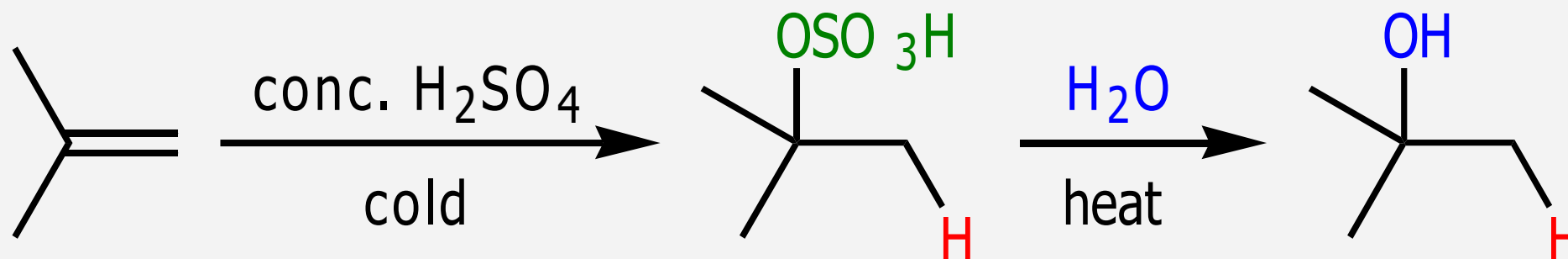
2. Addition of Sulfuric Acid to Alkenes



❖ Addition of $\text{H}-\text{OSO}_3\text{H}$ across a $\text{C}=\text{C}$ bond



Alcohols from Alkyl Hydrogen Sulfates (Hydrolysis)

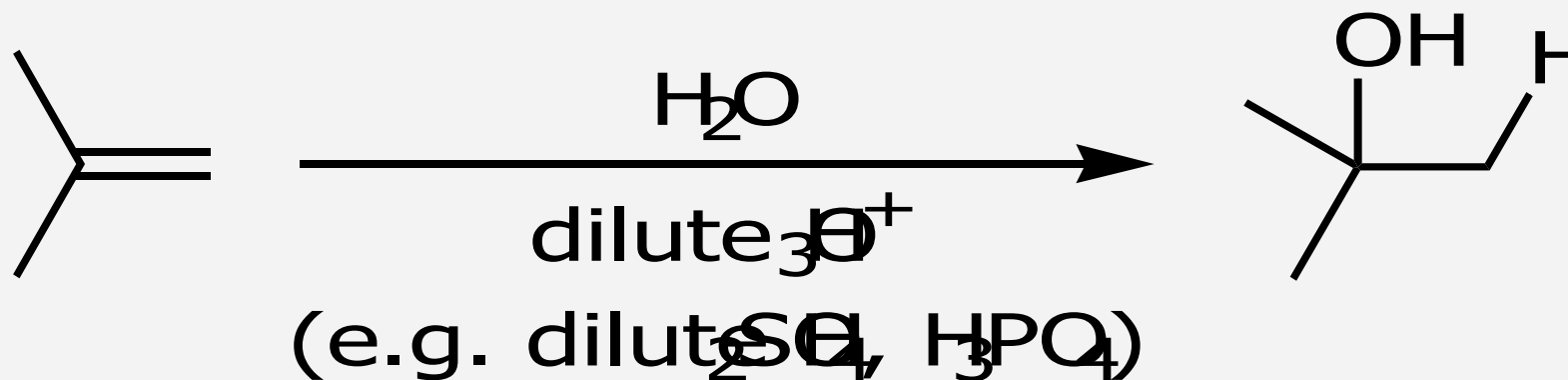


- ❖ The overall result of the addition of sulfuric acid to an alkene followed by hydrolysis is the Markovnikov addition of H- and -OH



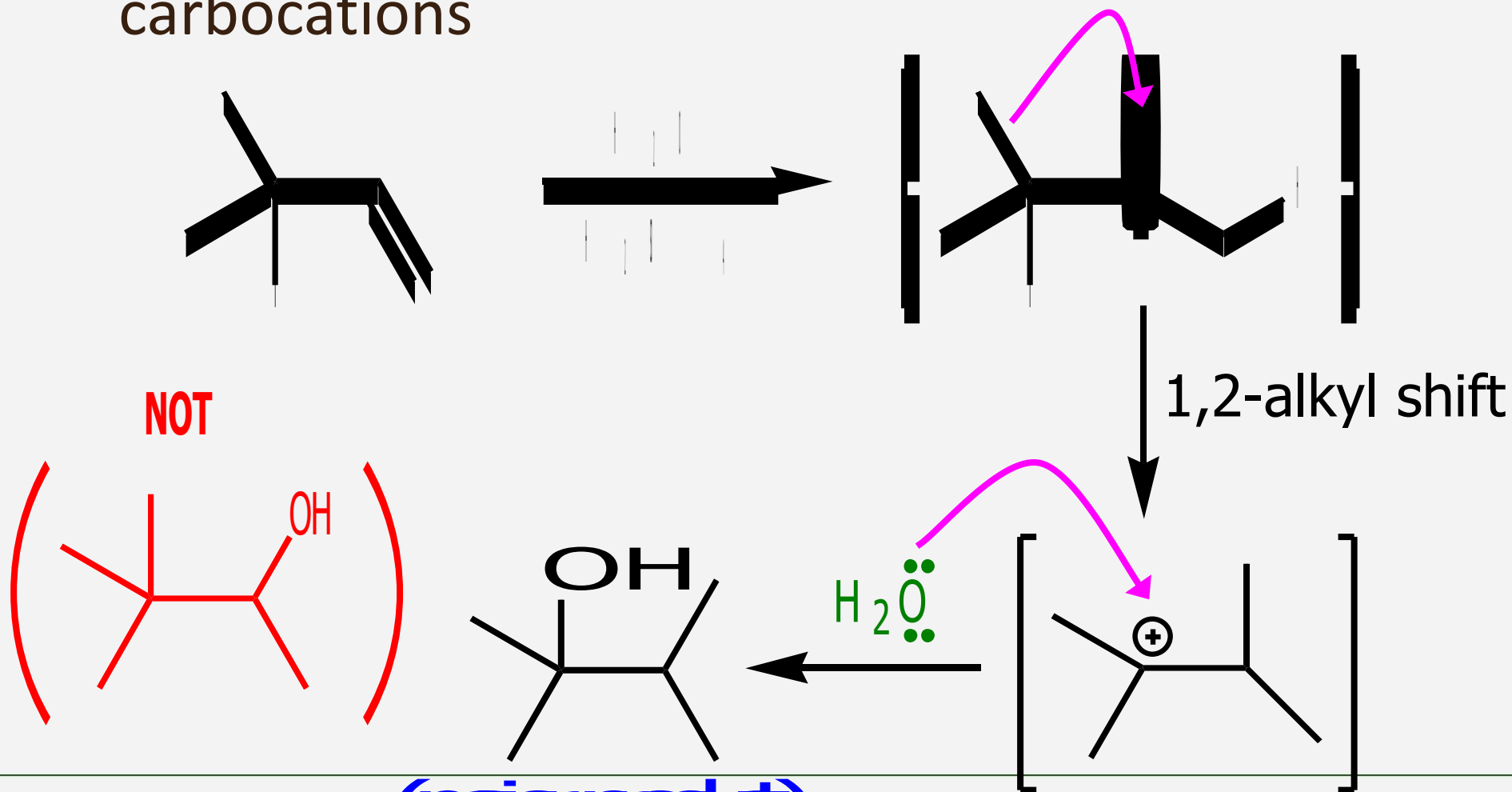
3. Addition of Water to Alkenes: Acid-Catalyzed Hydration

- ❖ Overall process
 - Addition of H–OH across a C=C bond
 - H^+ is the electrophile
 - Follow Markovnikov's rule



Rearrangements

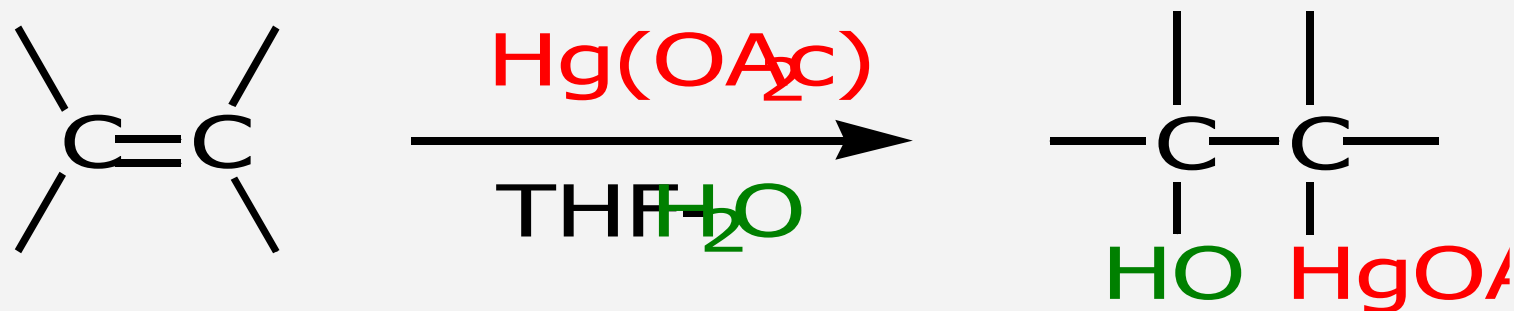
- ❖ Rearrangement can occur with certain carbocations



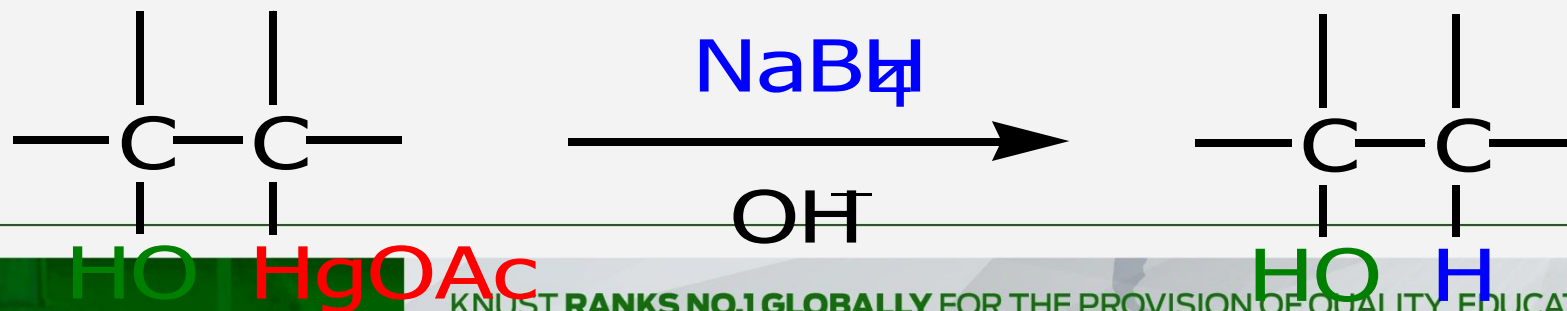
(major product)

4. Oxymercuration–Demercuration: Markovnikov Addition

❖ Step 1: Oxymercuration

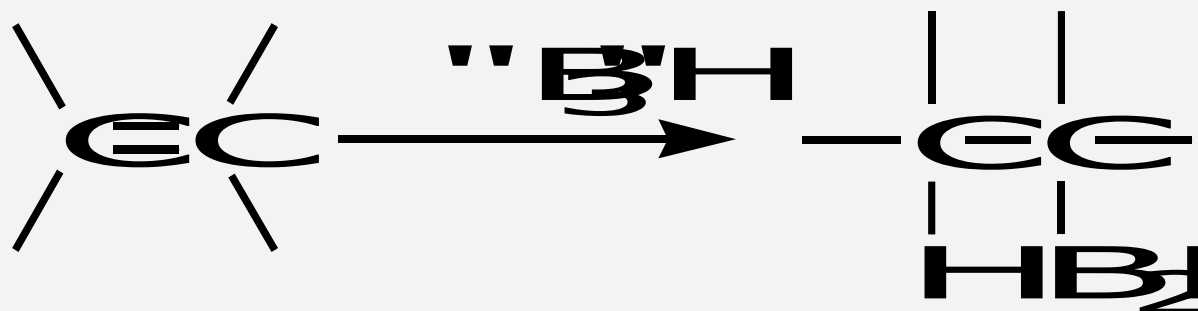


❖ Step 2: Demercuration



5. Hydroboration–Oxidation:

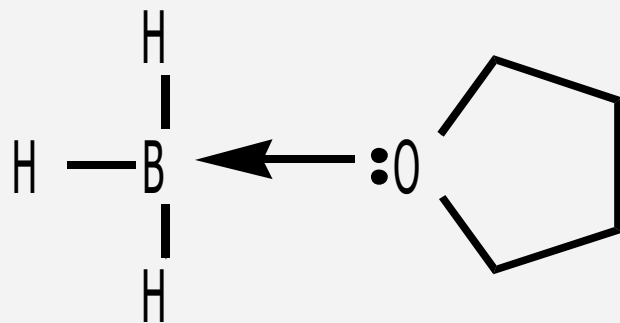
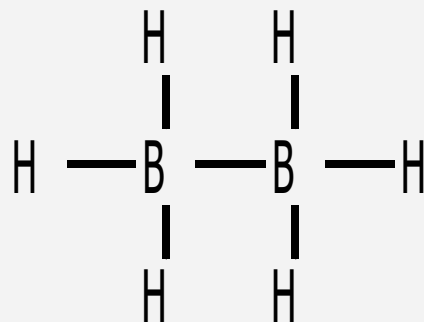
Alcohols from Alkenes through Anti-Markovnikov Syn Hydration



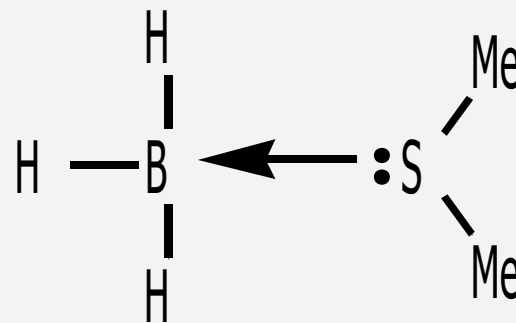
- ❖ Addition of $\text{H}-\text{BH}_2$ across a $\text{C}=\text{C}$ bond



- ❖ BH_3 exists as dimer B_2H_6 or complex with coordinative solvent

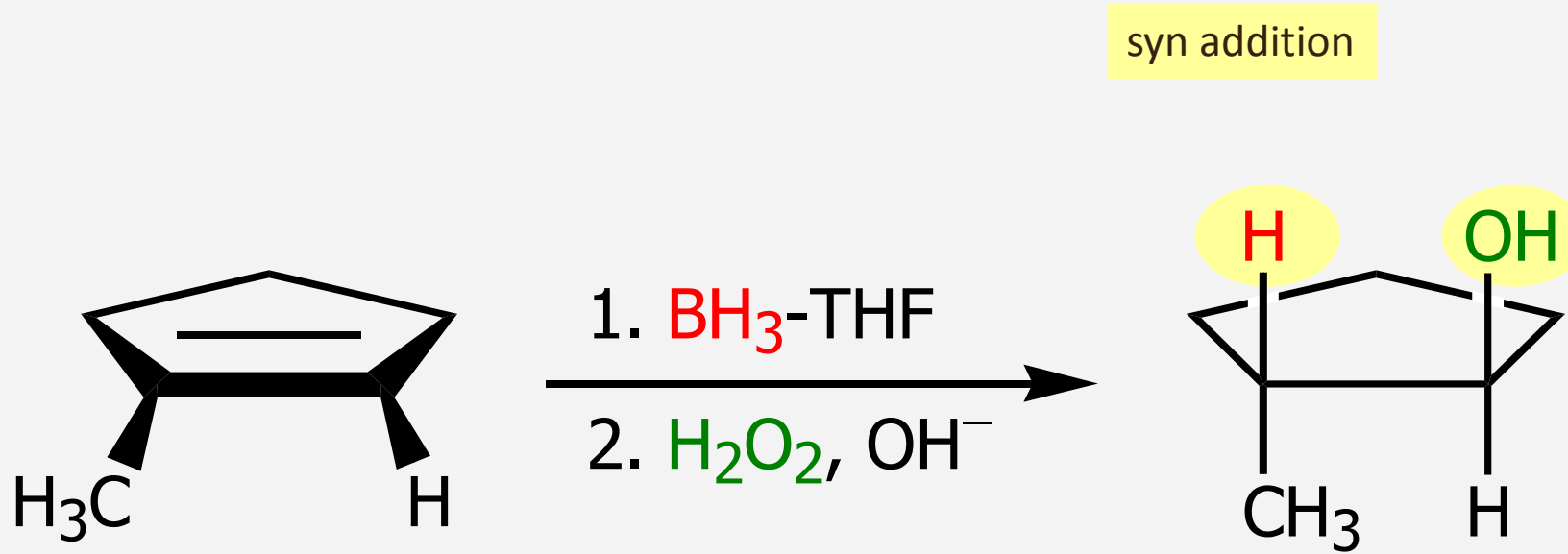


$(\text{BH}_3\text{-THF})$



$(\text{BH}_3\text{-DMS})$

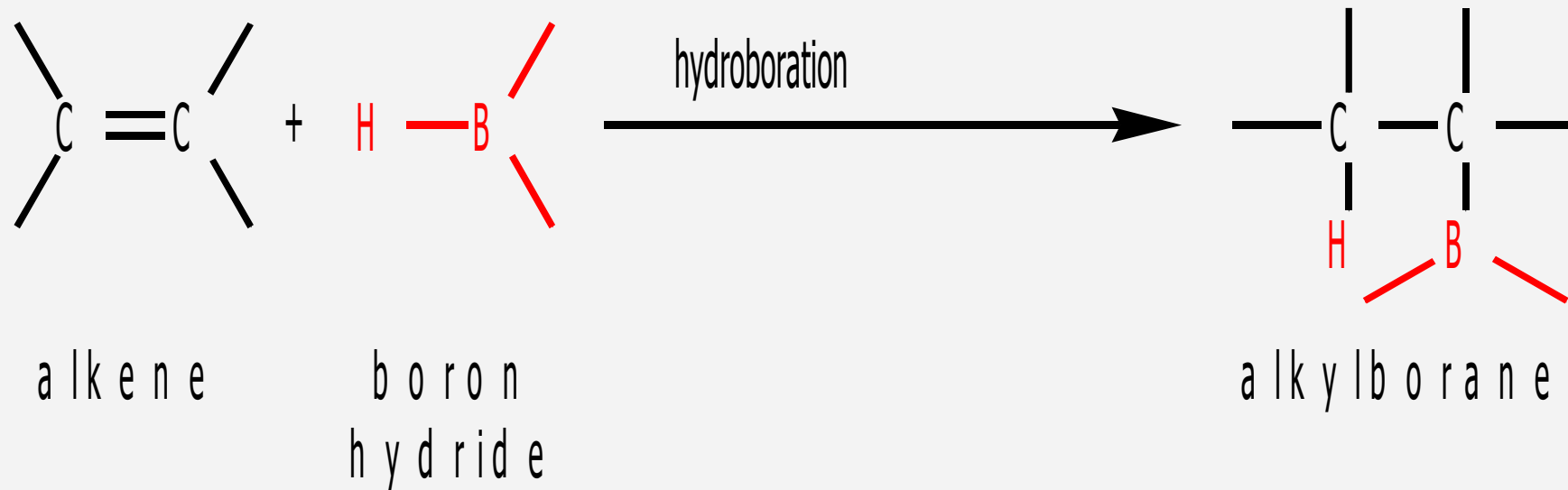




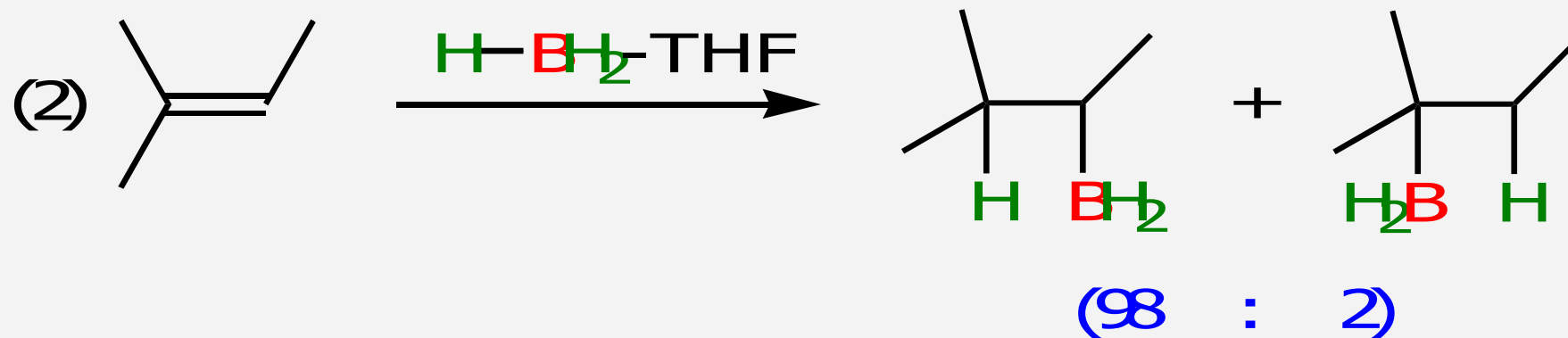
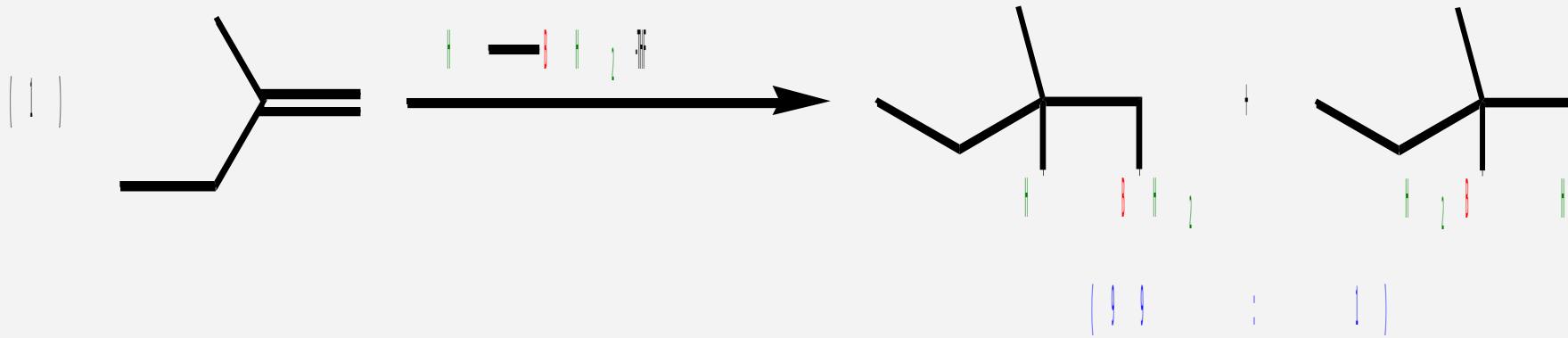
Anti-Markovnikov addition
of "H" & "OH"



6. Hydroboration: Synthesis of Alkylboranes

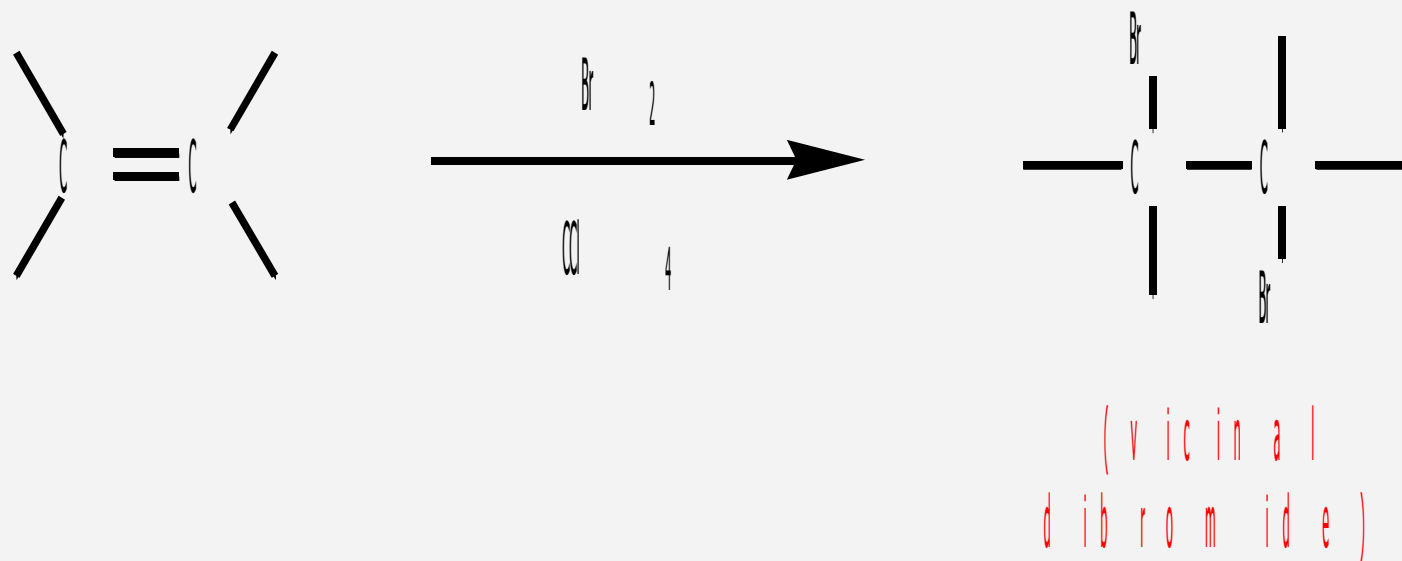


❖ Other examples

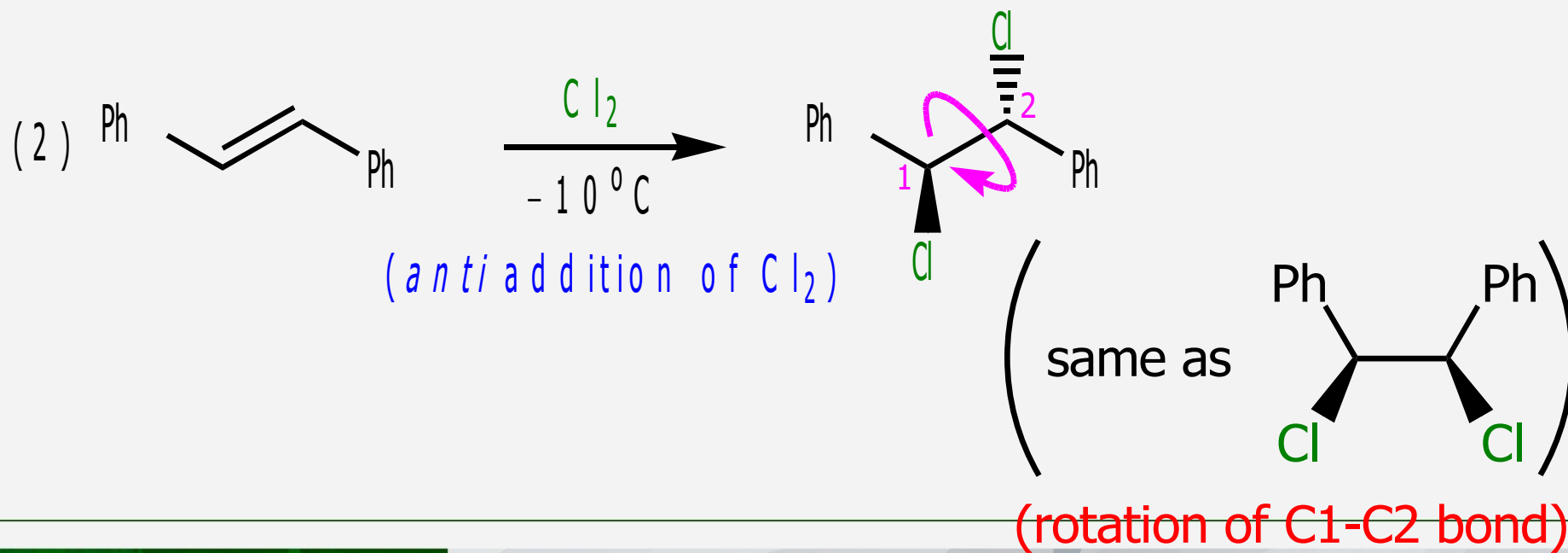
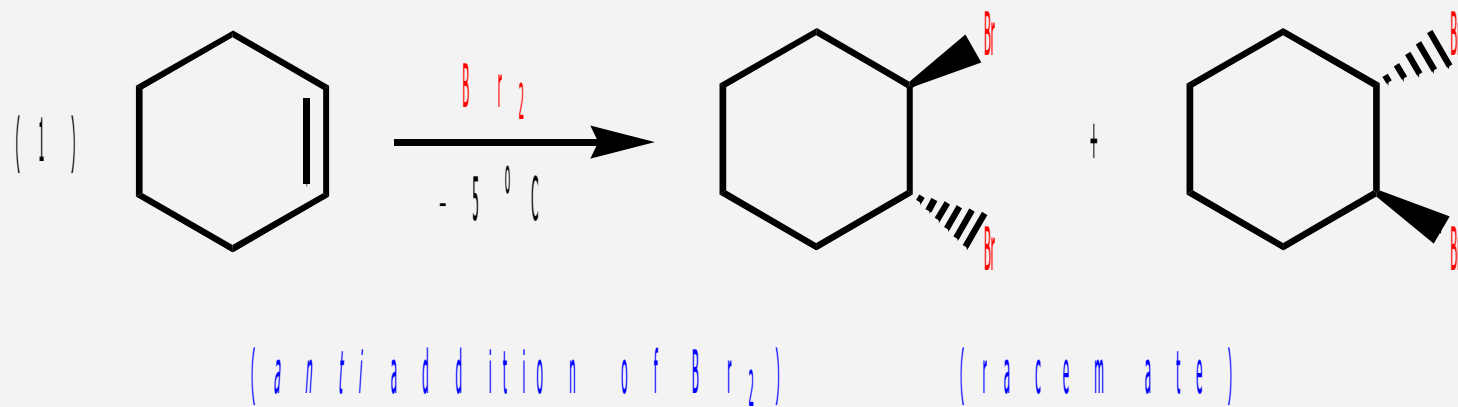


7. Electrophilic Addition of Bromine and Chlorine to Alkenes

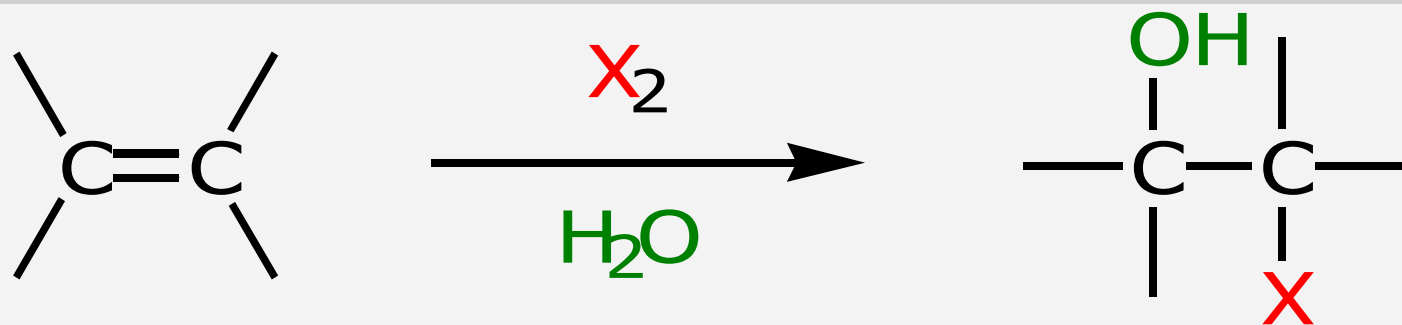
- ❖ Addition of X–X (X = Cl, Br) across a C=C bond



❖ Examples



8. Halohydrin Formation

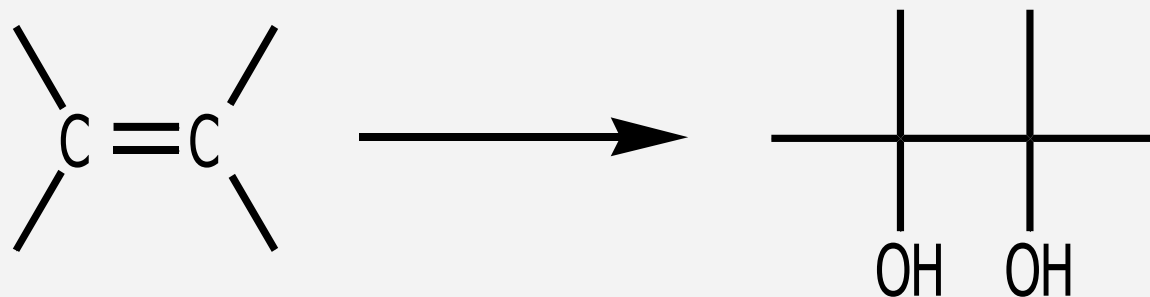


- ❖ Addition of -OH and -X ($\text{X} = \text{Cl}, \text{Br}$) across a $\text{C}=\text{C}$ bond
- ❖ X^+ is the electrophile
- ❖ Follow Markovnikov's rule



9. Oxidation of Alkenes: Syn 1,2-Dihydroxylation

- ❖ Overall: addition of 2 OH groups across a C=C bond

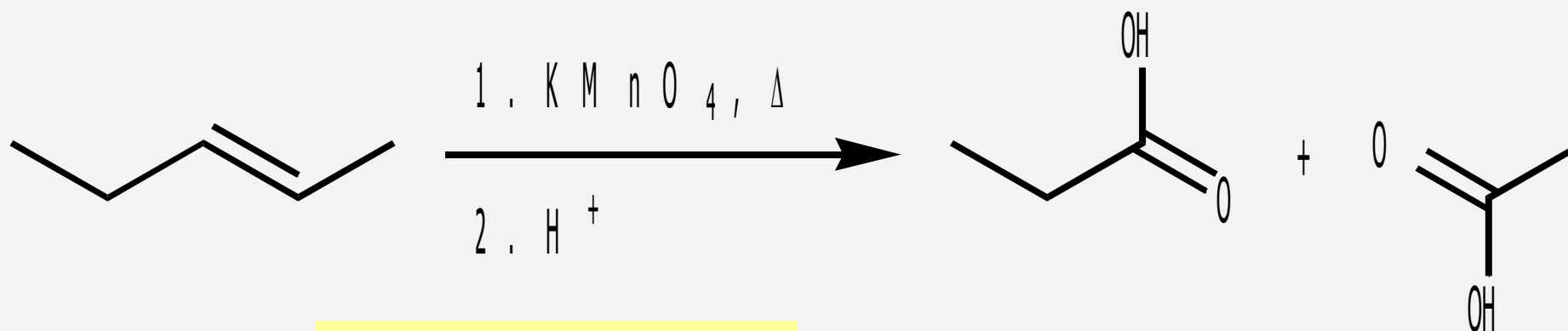


- ❖ Reagents: dilute KMnO_4 / $\text{OH}^\ominus$ / H_2O / cold or OsO_4 , pyridine then NaHSO_3 , H_2O



❖ Comparison of the two reagents

- KMnO_4 : usually lower yield and possibly side products due to over-oxidation



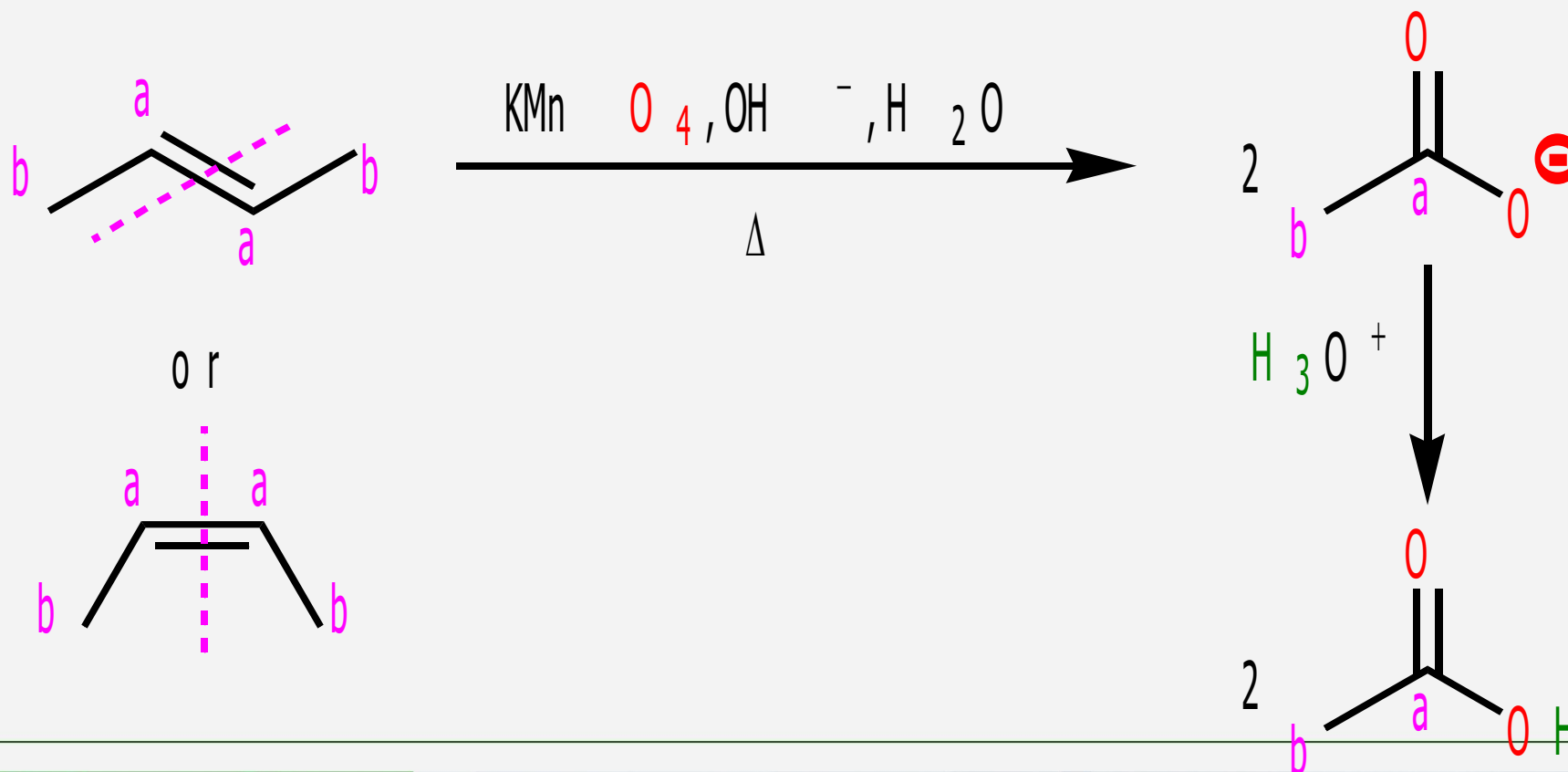
(oxidative cleavage of $\text{C}=\text{C}$)

- OsO_4 : usually much higher yield but OsO_4 is extremely toxic

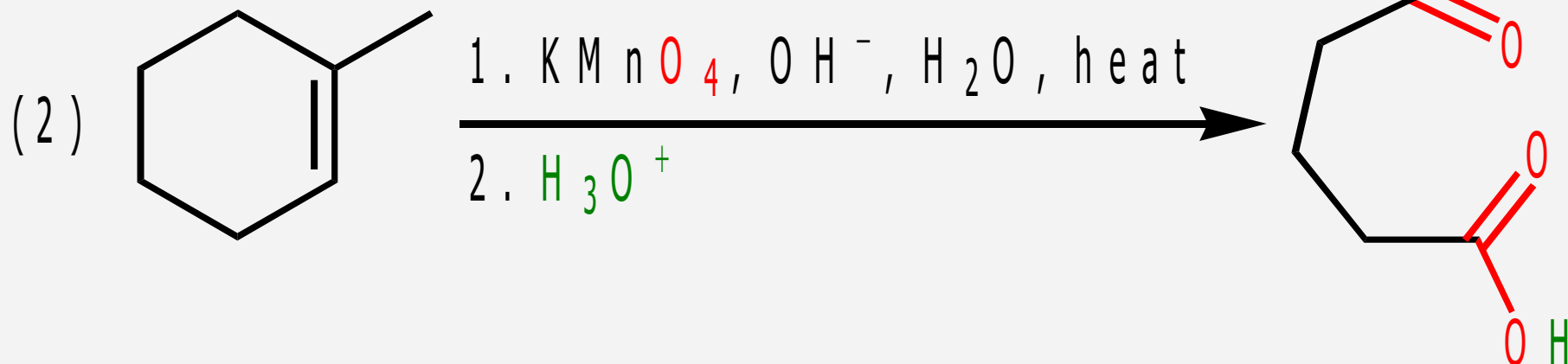
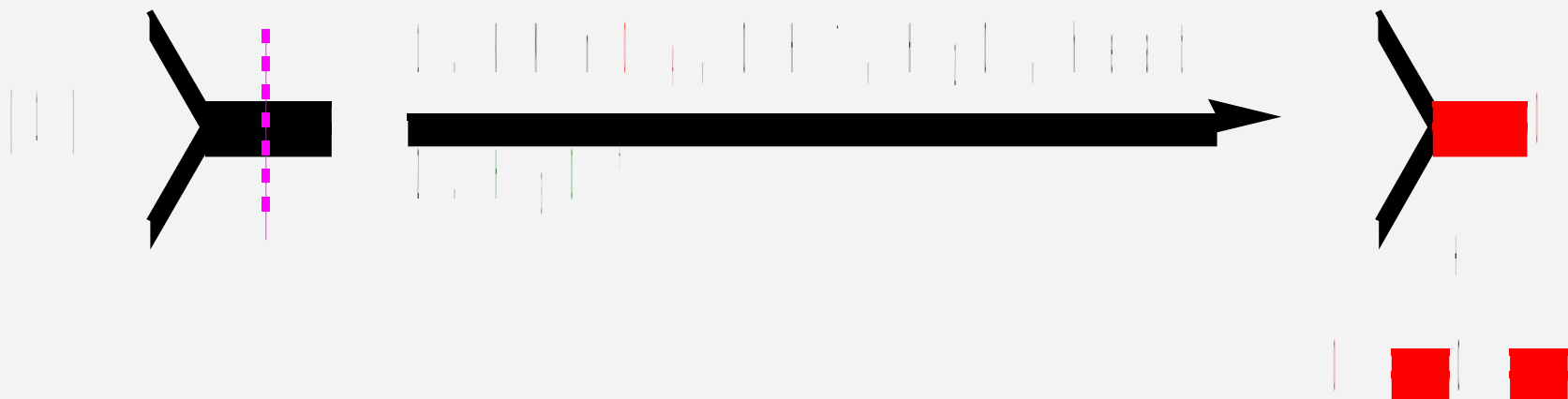


10. Oxidative Cleavage of Alkenes

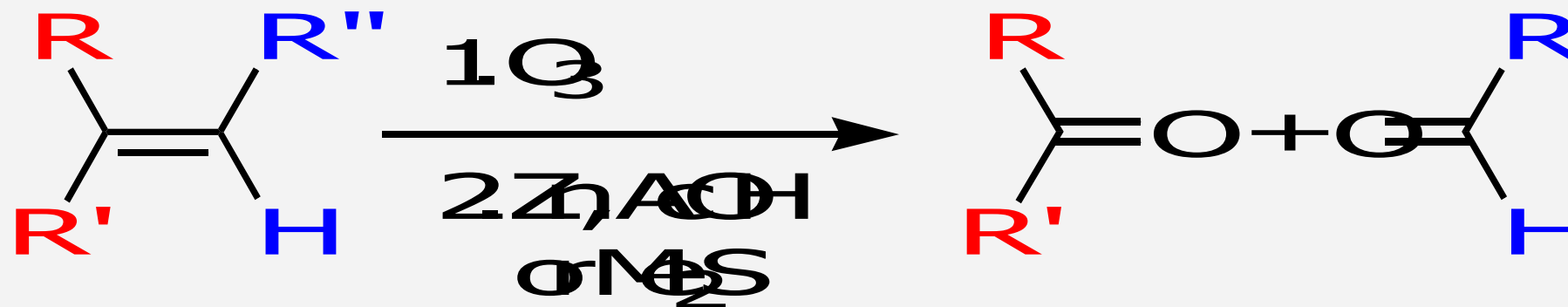
a) Cleavage with Hot Basic Potassium Permanganate



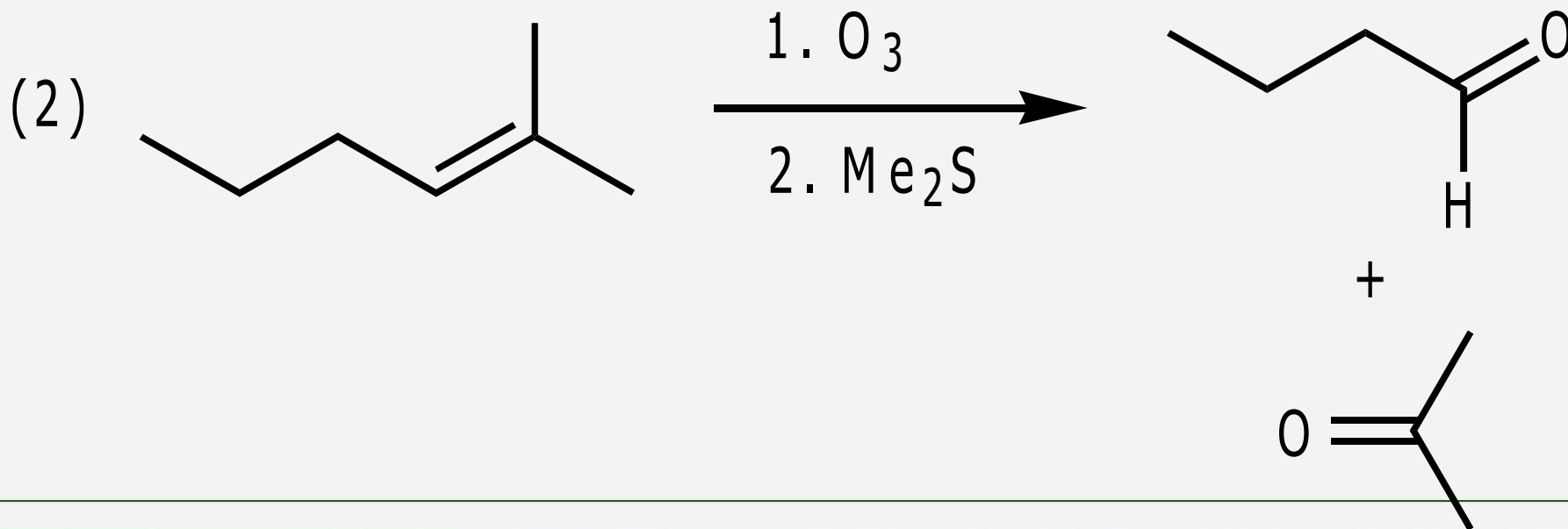
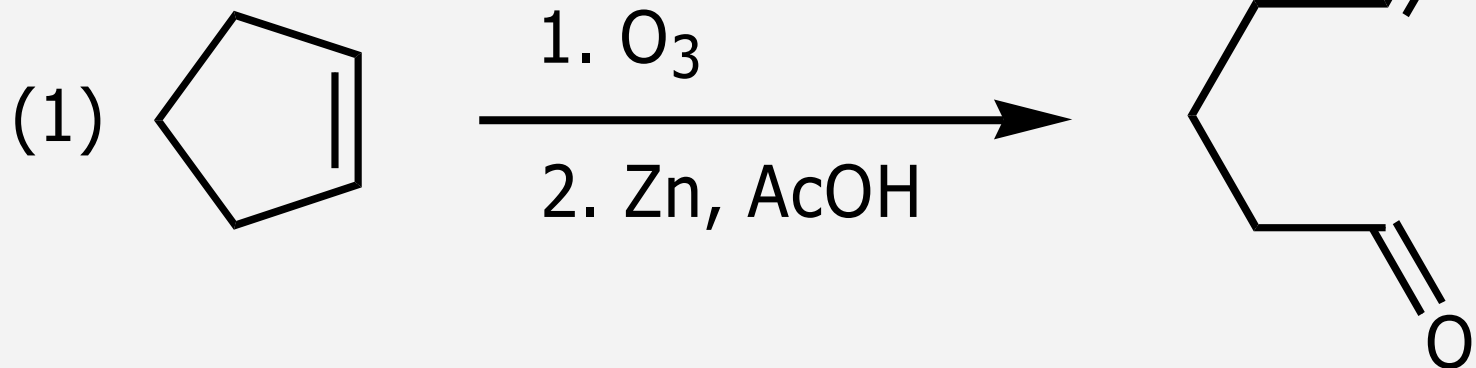
❖ Other examples



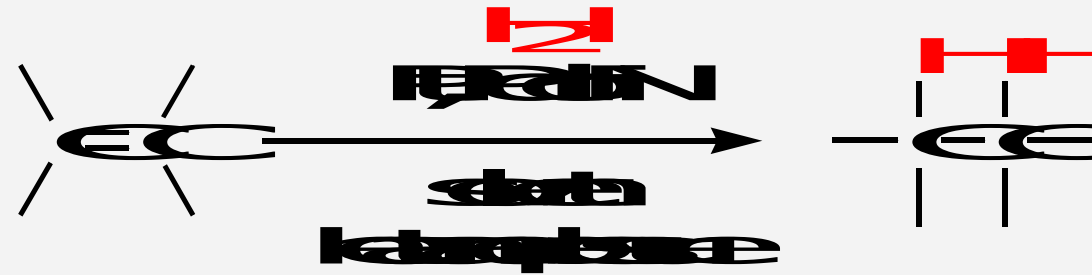
b) Cleavage with Ozone



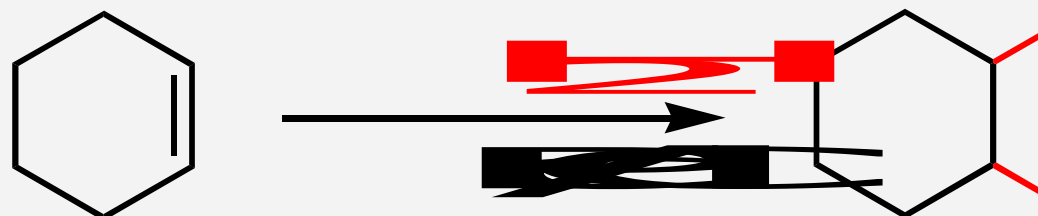
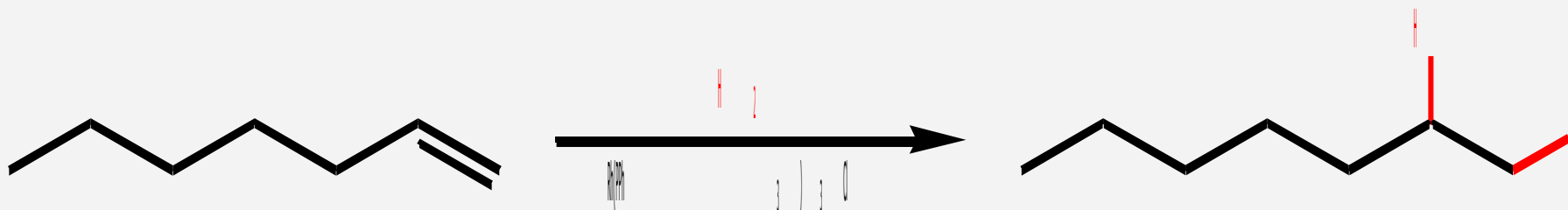
❖ Examples



11. CATALYTIC HYDROGENATION

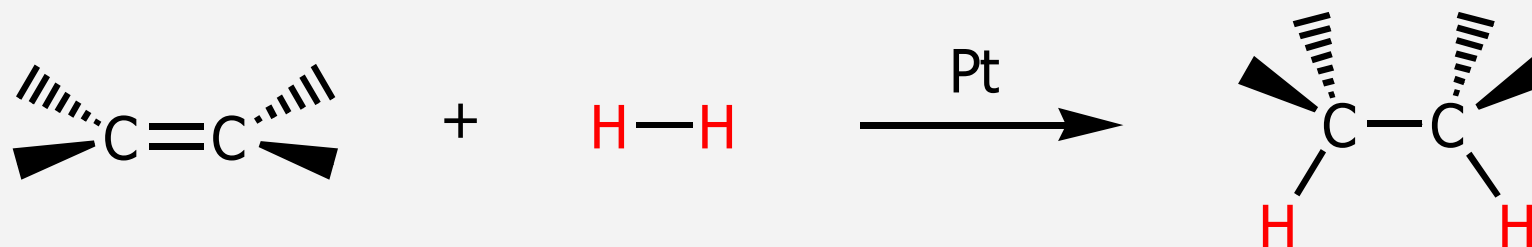
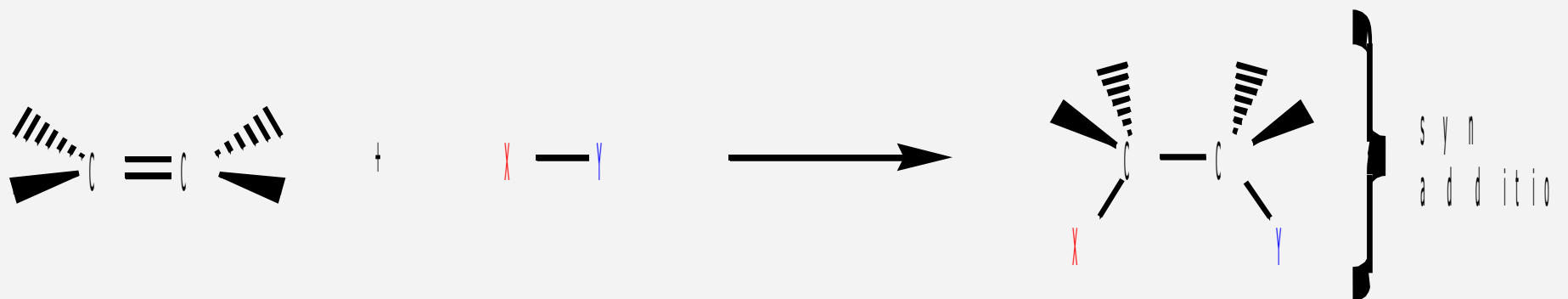


❖ Examples



Syn and Anti Additions

- ❖ An addition that places the parts of the reagent on the same side (or face) of the reactant is called **syn addition**



Catalytic hydrogenation is a **syn** addition.



- ❖ An **anti addition** places parts of the adding reagent on opposite faces of the reactant

